FEniCS Conference 2026 Book of Abstracts

FEniCS Conference 2026 Organizing Committee

10th Jun, 2026

Paris, 17-19 June 2026

Total abstracts: **67**

Wednesday - 17 June

Session 1: Software packages with later demonstrations

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14:00 - 14:15	Z3ST: A FEniCSx Framework for Multiscale Thermo-Mechanical Analysis with Automatic Differentiation	Giovanni Zullo
14:15 - 14:30	An Open-Source FEniCSx-Based Platform for Industrial Vibroacoustic Simulation	Antonio Baiano Svizzero
14:30 - 14:45	Automated generation of MITC shell elements in code_aster via FEniCSx symbolic engines	Joaquin Cornejo
14:45 - 15:00	FEniCSx in Metal Additive Manufacturing	Anton Evdokimov
15:00 - 15:15	Numerical investigation of urban heat island within variable porous urban domain	Luis Gerardo Gutierrez Ibarra
15:15 - 15:30	PhiFEM: an immersed boundary finite element method for geometries defined by a level-set	Michel Duprez

Session 2: Kernels, HPC deployment, and the wider FEniCS ecosystem

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16:15 - 16:30	Running after rounding errors: a posteriori error bounds of finite element kernels	Michal Habera
16:30 - 16:45	Distributing FEniCS with the EasyBuild/EESSI ecosystem: how to prepare a software package for distribution in HPC systems	Georgios Kafanas

Time	Contribution	Presenter
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17:00 - 17:15	Building a FEniCSx Ecosystem for Biomedical Research	Henrik Finsberg

Posters

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Shape optimization using PhiFEM	Raphaël Bulle
IMPACT - Modelling and optimisation of endless composite filament wound armour plates	Paul T. Kühner
What's new in dolfiny - high-level convenience wrappers for DOLFINx	Andreas Zilian
Adaptive edge element method for a quasilinear problem in electromagnetism with strong convergence for Gauss' law	Sanyang Liu
Flow distortion generation using topology optimization	Langlet Nathan
Effects of Pre-existing Defects on Helium Bubble Nucleation and Growth in Tungsten	Emna Frikha
Finite Element Modeling of Electrochemical Impedance Spectroscopy in All-Solid-State Batteries using FEniCSx	Théo Bermond
GPU kernels in DOLFINx	Chris Richardson
Numerical simulation of asphalt solar collector systems	Lucia Escudero Sartages
Integration of Externally Defined Constitutive Models into FEniCSx Using DOLFINx-External-Operator	Andrey Latyshev

Software demonstrations

Contribution	Presenter
CutFEMx: a cut finite element library for FEniCSx	Susanne Claus
Software demonstration (GPU)	Chris Richardson
Towards natural language-driven computational mechanics: A general LLM-integrated platform built on FEniCS	Guangjin Mou
Numerical investigation of urban heat island within variable porous urban domain	Luis Gerardo Gutierrez Ibarra
Live Demonstration of Z3ST: A FEniCSx Framework for Multiscale Thermo-Mechanical Analysis with Automatic Differentiation	Giovanni Zullo
Live Demonstration of an Open-Source Vibroacoustic GUI Built on FEniCSx	Antonio Baiano Svizzero
FEniCSx in Metal Additive Manufacturing	Anton Evdokimov
From Micro to Macro: Unsupervised FE2 Acceleration using ddfenicsx and micmacsfenicsx	Felipe Rocha
Integration of Externally Defined Constitutive Models into FEniCSx Using DOLFINx-External-Operator	Andrey Latyshev

Thursday - 18 June

Session 3: Multiphysics

Time	Contribution	Presenter
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9:15 - 9:30	Finite Element Modeling of Electrochemical Impedance Spectroscopy in All-Solid-State Batteries using FEniCSx	Théo Bermond
9:30 - 9:45	Multi-physics Modelling of Nuclear Fusion Reactor Components using FESTIM	James Dark
9:45 - 10:00	FESTIM v2.0: Upgraded framework for multi-species hydrogen transport	Remi Delaporte-Mathurin
10:00 - 10:15	Numerical quality factor statistics of multilayered superconducting radio-frequency cavity	Aaron Gobeyn
10:15 - 10:30	A fully monolithic formulation for multiphase fluid-structure interaction	Marc Hirschvogel

Session 4: Biomedical modelling and applications

Time	Contribution	Presenter
11:00 - 11:15	Nonlinear Inversion Framework - Magnetic Resonance Elastography (MRE)	Henrik Palme
11:15 - 11:30	In-silico drug delivery in the human brain	Cécile Daversin Catty
11:30 - 11:45	Data-integrated simulations of solute transport in the rodent brain using FEniCSx	Andreas Solheim
11:45 - 12:00	Poromechanics to Investigate the Impact of Mechanical Loading on Human Skin Micro-Circulation	Thomas Lavigne
12:00 - 12:15	Contact modeling using the Third Medium Contact (TMC) method: toward the application of a femoral stem insertion inside a bone cavity	Pierre Bichon
12:15 - 12:30	Inferring Human Intracranial CSF Absorption Sites via Inverse Modelling of Protein Transport	Marius Causemann

Session 5: Solid mechanics, contact, and material response

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14:00 - 14:15	Material model discovery with FEMU in FEniCSx	Saeid Ghouli
14:15 - 14:30	Finite-Pressure Anisotropic Hyperelasticity in FEniCSx: A Phenomenological Approach for Shock Physics	Paul Bouteiller
14:30 - 14:45	High-performance solid mechanics simulations in FEniCSx	Musa Choudhury
14:45 - 15:00	Flexibility Method for Contact Mechanics in FEniCSx. Finite Element Construction and Hierarchical Compression of Compliance Operators.	Yahya Boye
15:00 - 15:15	Computational Shakedown Analysis with FEniCSx: Conic Optimization for Large-Scale FEM	Lavakumar Veludandi
15:15 - 15:30	Convergence Study on Polycrystalline Materials	Donald Boyce

Session 6: Solver workflows, inverse modelling, and FEniCS ecosystem

Time	Contribution	Presenter
16:00 - 16:15	Variational Solvers for Irreversible Evolutionary Systems: Orchestration	Andrés A León Baldelli
16:15 - 16:30	Implementation of the TiNiest Tensor de Rham subcomplex on 3D Hybrid Meshes	Johnny Vogels
16:30 - 16:45	An implementation of the distributed two scales method for linear elasticity problems based on FEniCSx and PETSc	Salzman Alexis
16:45 - 17:00	Adjoint-accelerated inverse modelling of acoustic cross-talk in piezoelectric inkjet printheads	Javier Lorente-Macías
17:00 - 17:15	FEniCS Post-processing with ParaView	Louis Gombert
17:15 - 17:30	The Digital Math framework as a robust and general scientific basis for society - advancements in the Adaptive Euler breakthrough for aerodynamics and DigiMat educational program	Johan Jansson

Friday - 19 June

Session 7: Inverse problems, waves, and flow/transport

Time	Contribution	Presenter
9:15 - 9:30	Generalized Stokes problem with pressure Dirichlet boundary conditions for heat transport	Xavier Cartoixà
9:30 - 9:45	Electromagnetic Waveguide Input/Output Boundary Condition using SciFEM and FEniCSx	G. William Slade
9:45 - 10:00	Towards fully programmable inflatable panels	Ofir Mirkin
10:00 - 10:15	A Finite-Element Method for Fluctuating Navier–Stokes Equations	Dimitrios Gourzoulidis
10:15 - 10:30	Flow distortion generation using topology optimization	Langlet Nathan

Session 8: Engineering applications in structures, propulsion, and manufacturing

Time	Contribution	Presenter
11:00 - 11:15	Finite-Element Modeling of Complex Inelasticity Using FEniCSx: Paper and Thermoplastics	Johannes Neumann
11:15 - 11:30	Large scale random vibration analysis using FEniCS	Shubham Saurabh
11:30 - 11:45	femOHL: A FEniCSx-based Framework for Structural Integrity and Resilience Assessment of Overhead Power Transmission Infrastructure	Daria Mesbah
11:45 - 12:00	Multiphysics Eddy-Current Simulation for Efficient Hybrid-Electric Aircraft Propulsion	Arshad Fasiludeen
12:00 - 12:15	L-PBF Processes via FEniCS: A Finite Element Framework for Heat Modelling and Accelerated Predictive Workflows	José Ángel Bejarano Vázquez

CutFEMx: A Cut Finite Element Library for FEniCSx

Susanne Claus

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Submission type: Presentation

Presenter: Susanne Claus (ONERA)

CutFEMx is an extension to FEniCSx for cut finite element methods, released under the MIT license and available at https://github.com/sclaus2/CutFEMx. It extends DOLFINx with support for unfitted discretisations based on level set geometries, enabling the solution of partial differential equations on complex and evolving domains without meshing.

CutFEMx builds on two additional open-source components. The first, CutCells (MIT, available at https://github.com/sclaus2/CutCells), provides robust and efficient algorithms for computing intersections between level set functions and finite element cells of various types. It supports higher-order cutting up to polynomial degree (p=4) and generates both cut meshes for visualisation and quadrature rules on cut cells. The second component, runintgen (MIT, available at https://github.com/sclaus2/runintgen), extends FFCx with runtime quadrature kernel generation, allowing the integration of UFL forms over quadrature rules that vary between elements and are only defined at runtime, as required for cut cell integration.

The framework is designed with an emphasis on performance and modularity, combining cell-local operations, zero-copy data exchange, and compatibility with automatic differentiation workflows. The capabilities of CutFEMx are illustrated on representative examples, including Poisson, Stokes, and elasticity problems posed on implicitly defined domains. Application areas include shape and topology optimization, fluid-structure interaction, multi-material problems, and simulations involving evolving interfaces, where unfitted finite element methods provide a flexible alternative to mesh-conforming approaches.

Can we predict stress fields without models? Recent adventures of ddfenicsx for Data-driven Identification

Felipe Rocha

Université Paris-Est Créteil

Submission type: Presentation

Presenter: Felipe Rocha (MSME - Université Paris-Est Créteil)

Conservation laws are fundamental to physical modelling in electromagnetism, solid mechanics, and fluid dynamics. While the associated PDEs are universal, they require constitutive models – relating flux-gradient pairs (e.g., stress-strain) – to close the system. However, such models are not unique and often introduce assumptions that may bias predictions. Data-Driven Computational Mechanics (DDCM) [1] offers a paradigm shift: instead of assuming a constitutive law, it directly integrates discrete material data into the solution process. By reformulating the classical PDE problem as a mixed-integer minimisation, DDCM identifies mechanical states that satisfy equilibrium and kinematic compatibility while remaining closest to a dataset of stress-strain pairs. By operating directly on raw data, DDCM provides an alternative modeling route to both classical constitutive modeling and ML-based surrogates [2]. One interesting application of this approach lies in the intersection of experimental and computational mechanics, allowing stress field identification without explicitly postulating a parametric constitutive model, the so-called Data-Driven Identification (DDI) method [3]. This presentation shows recent advances of ddfenicsx [4], a library embedding DDCM within the FEniCSx ecosystem, preserving a familiar UFL-based workflow. Previous applications include coupled computational homogenisation [5], conservative mixed formulations for Darcy flow [6], and finite-strain problems [2], showcasing its versatility. In this talk, we focus on a new application of ddfenicsx to DDI using full-field displacement data, along with recent framework improvements: I) migration from legacy FEniCS to FEniCSx (0.10) ; ii) non-intrusive operation mode; iii) algorithmic enhancements such as the Accelerated Projection-to-Equilibrium [5] and Locally Convex Reconstruction [7] for noisy datasets and v) on-the-fly database generation [5].

References

[1] Kirchdoerfer, T., & Ortiz, M. (2016). Data-driven computational mechanics. Computer Methods in Applied Mechanics and Engineering, 304, 81-101. DOI: 10.1016/j.cma.2016.02.001

[2] Zlatic, M., Rocha, F., Stainier, L., & Canadija, M. (2024). Data-driven methods for computational mechanics: A fair comparison between neural networks based and model-free approaches. Computer Methods in Applied Mechanics and Engineering, 431, 117289. DOI: 10.1016/j.cma.2024.117289

[3] Leygue, A., Seghir, R., Réthoré, J., Tinnes, J.-P., & Coret, M. (2019). Non-parametric material state field extraction from full field measurements. Computational Mechanics, 64, 501-509. DOI: 10.1007/s00466-019-01725-z

[4] Rocha, F. (2023-). ddfenics: A FEniCS(x)-based (model-free) data-driven computational mechanics implementation [Software]. Zenodo. DOI: 10.5281/zenodo.7646226

[5] Rocha, F., Platzer, A., Leygue, A., & Stainier, L. (2025). On-the-fly adaptive sampling strategy for data-driven computational mechanics: Applications to computational homogenisation. Mechanics of Materials, 207, 105382. DOI: 10.1016/j.mechmat.2025.105382

[6] Rocha, F., Quinelato, T., & Stainier, L. (2024). Some experiences in mixed finite element formulations for (model-free) data-driven computational mechanics. In Proceedings of the 16ème Colloque National en Calcul de Structures (CSMA 2024). CNRS; CSMA. https://hal.science/hal-04611046

[7] He, Q., & Chen, J. S. (2020). A physics-constrained data-driven approach based on locally convex reconstruction for noisy database. Computer Methods in Applied Mechanics and Engineering, 363, 112791. DOI: 10.1016/j.cma.2019.112791

Z3ST: A FEniCSx Framework for Multiscale Thermo-Mechanical Analysis with Automatic Differentiation

Giovanni Zullo, Giovanni Nicodemo, Elisa Cappellari, Davide Pizzocri, Lelio Luzzi

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Submission type: Presentation

Presenter: Giovanni Zullo (Politecnico di Milano)

Z3ST is an open-source finite element framework built on FEniCSx. The framework provides an organized environment for coupled thermo-mechanical analysis with emphasis on differentiable constitutive modeling and inverse problem workflows, ranging from engineering-scale simulations to meso-scale phenomena. The framework couples thermal conduction, linear and nonlinear elasticity, hyperelasticity, phase-field fracture for material damage, macroscopic and crystal plasticity, and cluster dynamics through staggered schemes with adaptive relaxation. This demonstrates FEniCSx capability for production-level multiphysics codes across multiple length scales. Native support for UFL's differentiable architecture (Latyshev et al., 2025) underpins automated discovery of physics-informed constitutive laws through symbolic regression coupled with automatic differentiation (Flaschel et al., 2022). From the software engineering perspective, Z3ST provides complete simulation setup through external YAML configuration files for reproducibility while supporting Python-based customizable material laws. The framework includes 30+ verification cases ranging from analytical benchmarks to advanced applications. Z3ST demonstrates best practices for building domain-specific applications on FEniCSx, including effective use of quadrature function spaces for history variables, nonlinear solvers with adaptive relaxation, and seamless integration with visualization pipelines. Three relevant applications showcase Z3ST capabilities: (1) thermo-mechanical design-by-analysis applied to pressure vessel components, (2) microstructural analysis of over-pressurized bubbles in ceramics with intergranular fracture modeling, and (3) inverse parametric estimation for automated discovery of constitutive equations in irradiated materials (Frydrych, 2023). Code availability: https://github.com/giozu/z3st.

References

Alnaes, M. S., Logg, A., Olgaard, K. B., Rognes, M. E., & Wells, G. N. (2014). Unified form language: A domain-specific language for weak formulations of partial differential equations. ACM Transactions on Mathematical Software, 40(2), Article 9. DOI: 10.1145/2566630

Baratta, I. A., Dean, J. P., Dokken, J. S., Habera, M., Hale, J. S., Richardson, C. N., Rognes, M. E., Scroggs, M. W., Sime, N., & Wells, G. N. (2023). DOLFINx: The next generation FEniCS problem solving environment. Zenodo. DOI: 10.5281/zenodo.10447666

Flaschel, M., Kumar, S., & De Lorenzis, L. (2022). Discovering plasticity models without stress data. npj Computational Materials, 8, 91. DOI: 10.1038/s41524-022-00752-4

Frydrych, K. (2023). Modelling irradiation effects in metallic materials using the crystal plasticity theory - A review. Crystals, 13(5), 771. DOI: 10.3390/cryst13050771

Latyshev, A., et al. (2025). Expressing general constitutive models in FEniCSx using external operators and algorithmic automatic differentiation. Journal of Theoretical, Computational and Applied Mechanics. DOI: 10.46298/jtcam.14449

Scroggs, M. W., Dokken, J. S., Richardson, C. N., & Wells, G. N. (2022). Construction of arbitrary order finite element degree-of-freedom maps on polygonal and polyhedral cell meshes. ACM Transactions on Mathematical Software, 48(2), Article 18. DOI: 10.1145/3524456

An Open-Source FEniCSx-Based Platform for Industrial Vibroacoustic Simulation

Antonio Baiano Svizzero

Undabit

Submission type: Presentation

Presenter: Antonio Baiano Svizzero (Undabit)

An open-source vibroacoustic simulation platform built on FEniCSx is presented, designed as an alternative to established commercial solvers. The software targets a broad range of industrial applications, including automotive, aerospace and machinery noise control. The platform follows a three-layer architecture. The core layer relies on FEniCSx for the assembly and solution of variational problems. On top of it, a custom Python API provides domain-specific tools for vibroacoustic analysis: perfectly matched layers for modelling unbounded acoustic domains, a general shell formulation for thin-walled structures, a non-conformal coupling strategy between fluid and structural meshes that removes the need for conforming interfaces, and a mesh translator to import existing Nastran models directly into the FEniCSx workflow. Further modules are currently under development. The third layer is a graphical user interface based on PyQt6 and PyVista, which wraps the entire workflow into a desktop application that can be used without writing any Python code. We describe the architecture of the platform and each of its tools, then illustrate the current capabilities of the prototype through representative vibroacoustic benchmark cases and compare the results with those obtained from commercial solvers. Although the platform is still at an early stage, its modular design makes it easy to extend with new physics and formulations. By combining FEniCSx with industry-oriented tools and a user-friendly interface, this project aims to make high-fidelity vibroacoustic simulation more accessible to both researchers and practising engineers.

Automated generation of MITC shell elements in code_aster via FEniCSx symbolic engines

A. Bouaziz, **J. Cornejo**, N. Tardieu, V. Le Corvec

EDF Lab Paris-Saclay

Submission type: Presentation

Presenter: Joaquin Cornejo (EDF Lab Paris Saclay)

This project introduces a novel framework that bridges the symbolic automation of FEniCSx with Code_Aster. While traditional element development in Code_Aster requires tedious manual derivation and Fortran implementation, this approach takes advantage of FEniCSx to automatically generate optimized C-code kernels directly from variational energy formulations. This paradigm shift not only accelerates development cycles but also ensures mathematical exactness by eliminating manual differentiation.

The core of this work details the integration of a Reissner-Mindlin shell formulation using Mixed Interpolation of Tensorial Components (MITC) to inherently bypass shear locking. We address the non-trivial architectural challenges of mapping FEniCSx-generated kernels onto Code_Aster's Fortran environment, specifically regarding disparate nodal numbering conventions and the management of non-standard degree of freedom.

A significant breakthrough was achieved in the treatment of Nédélec-type elements. To maintain compatibility with Code_Aster's assembler, we implemented a local static condensation strategy. By eliminating Nédélec degrees of freedom at the element level, we successfully recovered a nodal-only interface that matches the efficiency and structure of Code_Aster solvers while preserving the benefits of the MITC formulation.

Current results demonstrate perfect agreement with established benchmarks in linear elasticity. However, the true potential of this framework lies in its future: we are working on integrating MFront and JAX to handle non-linear constitutive laws through automatic differentiation and code generation. This work paves the way for a new shell element in Code_Aster, one that combines the flexibility of modern research tools with the rigorous safety standards required for nuclear industry simulations.

References

Hale JS, Brunetti M, Bordas SP, Maurini C. Simple and extensible plate and shell finite element models through automatic code generation tools. Computers & Structures. 2018 Oct 15;209:163-81.

Latyshev A, Bleyer J, Maurini C, Hale J. Expressing general constitutive models in FEniCSx using external operators and algorithmic automatic differentiation. Journal of Theoretical, Computational and Applied Mechanics. 2025 Sep 22.

FEniCSx in Metal Additive Manufacturing

Anton Evdokimov

heatshape GmbH

Submission type: Presentation

Presenter: Anton Evdokimov (heatshape GmbH)

The talk covers the application of FEniCSx to the simulation of a metal 3D printing process, namely wire arc additive manufacturing (WAAM). The focus will be on the definition of moving heat sources and element activation strategies. It is demonstrated how simulation can help analyze temperatures, distortions, and microstructural evolution during metal printing, thus improving the quality and efficiency of metal additive manufacturing.

Numerical investigation of urban heat island within variable porous urban domain

Luis Gerardo Gutierrez Ibarra, Néstor García Chan, Juan Antonio Licea Salazar

University of Guadalajara

Submission type: Presentation

Presenter: Luis Gerardo Gutierrez Ibarra (University of Guadalajara, Mexico)

This paper presents a numerical study of the Urban Heat Island (UHI), defined as the temperature differential between urban and rural or suburban areas. Urban morphology is modeled as a variable porous medium, where buildings and streets represent the solid and fluid phases, respectively, enabling the meshing of large-scale cities. UHI intensity is measured through air temperature, coupled with an energy balance to account for the heating of streets and buildings due to solar radiation. To achieve this, a two-dimensional vertical dynamic model based on partial differential equations was employed, utilizing a Darcy-Forchheimer-Brinkman type equation to obtain the wind velocity field. Furthermore, a thermal exchange model is used between air and building temperatures; additionally, a surface energy balance equation is coupled, considering net radiation and the physical properties of the urban surface.Estimating UHI intensity at a macroscale can be instrumental in decision-making processes regarding urban planning, benefiting the environment and urban residents in both economic and health terms. Moreover, it provides a framework to explore UHI mitigation strategies. This study considers a domain of 11 km x 800 m with variable porosity throughout, assuming a concentric porosity distribution (lower in the center and higher in the suburbs), utilizing meteorological data for solar radiation. For the numerical solution, the Taylor-Hood finite element method and an IPCS scheme with BDF2 discretization and stabilizers were implemented using the FEniCSx software. Finally, various numerical experiments are presented to visualize the effects of porosity, wind, and radiation on UHI intensity.

PhiFEM: an immersed boundary finite element method for geometries defined by a level-set

Raphaël Bulle[1], **Michel Duprez**[1], Vanessa Lleras[2], Alexei Lozinski[3], Killian Vuillemot[2]

[1] Inria
[2] Université de Montpellier
[3] Université de Besançon

Submission type: Presentation

Presenter: Michel Duprez (Inria)

In this talk, we will present a fictitious domain finite element method called φ-FEM, well suited for elliptic problems posed in a domain given by a level-set function without requiring a mesh fitting the boundary. Unlike other recent fictitious domain-type methods (XFEM, Cut-FEM), our approach does not need any non-standard numerical integration, neither on the cut mesh elements nor on the actual boundary. We shall present the proofs of optimal convergence of our methods on the example of Poisson equation using Lagrange finite elements of any order. Furthermore, we highlight the flexibility and efficiency of our method on different problems (crack, interface, Stokes, heat equation,...). We will also give numerical tests illustrating the optimal convergence of our methods and discuss the conditioning of resulting linear systems and the robustness with respect to the geometry.

References

[1] R. Becker, R. Bulle, M. Duprez, V. Lleras. Residual-based a posteriori error estimates with boundary correction for φ-FEM, in preparation.

[2] M. Duprez, A. Lozinski, V. Lleras, V. Vigon, K. Vuillemot. A finite difference scheme with an optimal convergence for elliptic PDEs on domains defined by a levelset function, Journal of Scientific Computing, 104 (23), 2025.

[3] M. Duprez, V. Lleras, A. Lozinski. A new φ-FEM approach for problems with natural boundary conditions. Numerical Methods for Partial Differential Equations, 39(1):281-303, 2023.

[4] M. Duprez M, V. Lleras and A. Lozinski. φ-FEM: an optimally convergent and easily imple-

mentable immersed boundary method for particulate flows and Stokes equations. ESAIM: Mathematical Modelling and Numerical Analysis, 1;57(3):1111-42, 2023.

[5] S. Cotin, M. Duprez, V. Lleras, A. Lozinski and K. Vuillemot. φ-FEM: An Efficient Simulation

Tool Using Simple Meshes for Problems in Structure Mechanics and Heat Transfer. Partition of Unity Methods, 19:191-216, 2023.

[6] M.Duprez, V. Lleras, A. Lozinski, K. Vuillemot. φ-FEM for the heat equation: optimal convergence on unfitted meshes in space, Comptes Rendus. Mathématique, Tome 361, 1699:1710, 2023.

[7] M. Duprez, A. Lozinski. φ-FEM: a finite element method on domains defined by level-sets. SIAM Journal on Numerical Analysis, 58(2):1008-28, 2020.

Fast Finite Element Kernels for Modern Computing Architectures

Joseph P. Dean[1], Igor A. Baratta[2], Chris N. Richardson[1], Garth N. Wells[1]

[1] University of Cambridge
[2] NVIDIA

Submission type: Presentation

Presenter: Joseph Dean (University of Cambridge)

Accurate simulation of increasingly complex systems requires numerical methods that can fully exploit modern HPC hardware. This demands algorithms that minimise memory traffic, make effective use of memory hierarchies, and exploit vectorisation. In this work, we present high-order matrix-free finite element kernels designed for modern CPU and GPU architectures and examine the key implementation challenges involved in achieving high performance. Using parameterisable algorithms that allow tuning for specific hardware, we explore how the balance of memory pressure can be shifted between different hardware components. We use profiling and benchmarking to investigate the effects on cache reuse, memory bandwidth, and register pressure. Performance results are presented for a range of hardware, including Intel Sapphire Rapids and NVIDIA Grace CPUs, as well as AMD MI300X and NVIDIA GH200 GPUs. In addition, we show that low-order methods can remain more competitive than is sometimes assumed.

Running after rounding errors: a posteriori error bounds of finite element kernels

Michal Habera[1], Paul T. Kühner[1], Matteo Croci[2], Andreas Zilian[1]

[1] University of Luxembourg
[2] Basque Center for Applied Mathematics

Submission type: Presentation

Presenter: Michal Habera (University of Luxembourg)

Rounding errors in finite element computations can lead to significant inaccuracies, stalled convergence, and incorrect results. Moreover, the effects of rounding errors accumulated within automatically generated and compiled kernels are difficult to analyze a priori. In this contribution, we use an a posteriori technique called Running Error Analysis (REA), where a forward error bound is computed concurrently with the value.

We present the implementation of REA for the FEniCS Form Compiler (FFCx), based on a C++ backend for generating type-generic templated kernels over a custom arithmetic type that tracks both the value and its error bound. As an example, we demonstrate REA for detecting catastrophic cancellation in the assembly of a Neo-Hookean hyperelastic model in the small deformation regime. We also show that REA offers small performance overhead compared to higher- or multi-precision kernel evaluation (e.g., `std::float128` or `mpfr_t`). Applications of this work include: robust reduced-precision computations in embedded systems, numerical debugging of new, possibly ill-conditioned or unstable PDE formulations, and guiding the design of mixed-precision kernels.

The functionality is available for arbitrary UFL forms supported by FFCx and is implemented in the `dolfiny` framework.

Distributing FEniCS with the EasyBuild/EESSI ecosystem: how to prepare a software package for distribution in HPC systems

Georgios Kafanas, Hassan Md Jahid, Julien Schleich

University of Luxembourg

Submission type: Presentation

Presenter: Georgios Kafanas (University of Luxembourg)

Very often getting a piece of scientific software working properly is the greatest impediment in starting using a new software system. Modern scientific software like FEniCS is usually composed of a large collection of software packages, and each of these package was dependencies of their own. Furthermore, the deployment targets of FEniCS are equally varied, with HPC centers hosting a variety of CPU architectures and models, and an equally varied amount of peripherals, like storage, accelerators, and networking. To manage this complexity, source code software management systems like EasyBuild and Spack have been developed. The process of distribution is now automated to a significant degree as well, with services like EESSI, which offers streaming access to a uniform and platform-optimized set of software modules and is deployed in all EuroHPC Federated centers.

Software distribution system operated in distinct phases, to compile source code, and then test, and deploy binaries. This talk will describe the deployment of FEniCS in the EasyBuild/EESSI ecosystem, that relies in EasyBuild for source code compilation, ReFrame for testing, and EESSI for the streaming of platform-optimized binaries. Emphasis will be placed on the subtle design modification required so that the software build and test systems follow the conventions in place in the EasyBuild/EESSI ecosystem. Although the presentation focuses on FEniCS and EESSI, the underlying principle and practices presented are applicable to the build and test systems of any piece of software that aims to be distributed with a modern software deployment system.

References

Karakasis, V., Manitaras, T., Rusu, V. H., Sarmiento-Pérez, R., Bignamini, C., Kraushaar, M., ... Tomko, K. (2020). Enabling continuous testing of HPC systems using ReFrame. In Tools and Techniques for High Performance Computing (pp. 49-68). DOI: 10.1007/978-3-030-44728-1_3

Geimer, M., Hoste, K., & McLay, R. (2014, November). Modern Scientific Software Management Using EasyBuild and Lmod. 2014 First International Workshop on HPC User Support Tools. DOI: 10.1109/hust.2014.8

Gamblin, T., LeGendre, M., Collette, M. R., Lee, G. L., Moody, A., de Supinski, B. R., & Futral, S. (2015, November). The Spack package manager: bringing order to HPC software chaos. Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis, 1-12. DOI: 10.1145/2807591.2807623

Dolstra, E. (2006). The Purely Functional Software Deployment Model (Doctoral dissertation, Utrecht, The Netherlands). Retrieved from https://researchr.org/publication/Dolstra2006

Dolstra, E., Löh, A., & Pierron, N. (2010). NixOS: a purely functional Linux distribution. Journal of Functional Programming, 20(5-6), 577-615. DOI: 10.1017/s0956796810000195

Malka, J., Zacchiroli, S., & Zimmermann, T. (2025, April). Does Functional Package Management Enable Reproducible Builds at Scale? Yes. 2025 IEEE/ACM 22nd International Conference on Mining Software Repositories (MSR), 775-787. DOI: 10.1109/msr66628.2025.00115

Evolving FEniCS: The Extension Ecosystem at Simula Scientific Computing

Jørgen S. Dokken, Henrik N.T. Finsberg

Simula Research Laboratory

Submission type: Presentation

Presenter: Jørgen S. Dokken (Simula Research Laboratory)

The FEniCS project provides the foundational algorithms for finite element computations, which in turn has organically cultivated a rich ecosystem of software extensions. This talk will highlight several key extensions developed and maintained by Simula Research Laboratory. Specifically, I will introduce SciFem, a utility toolbox for DOLFINx; io4dolfinx, a modern IO-library; and FEniCSx_ii, a library designed for non-local operators. The presentation will conclude with a look ahead at frameworks currently under active prototyping, including DXA (dolfinx-adjoint), FEniCSx-Jax and networks_fenicsx.

Building a FEniCSx Ecosystem for Biomedical Research

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Submission type: Presentation

Presenter: Henrik Finsberg (Simula Research Laboratory)

Translating physiological processes into predictive models requires robust and performant software pipelines. While DOLFINx offers a high-performance finite element backend, applying it to complex biomedical domains means bridging the gap between raw clinical data, specialized physical models, and scalable solvers. In this talk, we present a modular ecosystem of open-source FEniCSx-based tools designed to streamline workflows in brain and cardiac modelling.

For brain modelling, a major challenge is seamlessly integrating clinical imaging data into the finite element pipeline. We introduce MRI-toolkit, a specialized tool for converting MRI sequences into simulation-ready inputs, such as spatially varying concentration maps. We also discuss efficient strategies using wildmeshing and pyvista to generate high-quality computational meshes directly from medical image segmentations, bridging voxel data and FEniCSx-ready unstructured grids.

In cardiac modelling, we present a comprehensive suite of interoperable packages targeting various scales and physics. At the cellular level, gotranx facilitates the generation of ODE code for single-cell models. At the tissue and organ level, fenicsx-beat provides robust solvers for cardiac electrophysiology, while fenicsx-pulse simulates nonlinear cardiac mechanics. To support these models, we introduce cardiac-geometriesx for generating parametrised cardiac geometries, fenicsx-ldrb for assigning realistic myocardial fiber architectures, and circulation for coupling 3D finite element models to 0D lumped-parameter networks.

Finally, we share critical lessons learned from developing and maintaining this scientific software ecosystem, discussing our approaches to automated testing, package distribution (PyPI/Conda/Spack), and creating user-friendly documentation.

References

[1] MRI-toolkit, https://github.com/scientificcomputing/mri-toolkit

[2] PyVista, https://github.com/pyvista/pyvista

[3] gotranx, https://github.com/finsberg/gotranx

[4] ap features, https://github.com/ComputationalPhysiology/ap_features

[5] fenicsx-beat, https://github.com/finsberg/fenicsx-beat

[6] fenicsx-pulse, https://github.com/finsberg/fenicsx-pulse

[7] cardiac-geometriesx, https://github.com/ComputationalPhysiology/cardiac-geometriesx

[8] fenicsx-ldrb, https://github.com/finsberg/fenicsx-ldrb

[9] circulation, https://github.com/ComputationalPhysiology/circulation

Developing numerical tools to improve the diagnosis of peripheral artery disease

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University of Leeds

Submission type: Poster

Presenter: Luke J Barratt (University of Leeds)

Peripheral artery disease (PAD) is an age-related condition present in >10% of adults over 65 years and is characterised by atherosclerotic narrowing of arteries classically affecting the legs [1]. This reduces oxygen delivery to skeletal muscle and leads to pain during walking (intermittent claudication, IC) [2]. Early diagnosis of PAD is key to reducing risk of cardiovascular events; a risk that is two-fold greater in PAD than in those with stable coronary artery disease [3]. The gold-standard diagnostic method for PAD is the ankle-brachial index (ABI; the ratio of leg to arm systolic blood pressure), sometimes supplemented with ultrasound, MRI or CT [4]. However, traditional diagnostic methods do not always detect PAD, particularly in patients with calcified arteries or atypical presentations often occurring in women. As a result, PAD frequently remains underdiagnosed and undertreated compared with other cardiovascular conditions [5]. To address the limitations of current diagnostic methods, we are developing numerical methods to improve risk prediction algorithms by assessing factors local to the occluded limbs. Three mathematical models are being developed that can assess: (1) blood flow in the large arteries using 1D Navier-Stokes for flow in compliant tubes [6], (2) microvascular flow using Poiseuille's Law in networks [7], (3) dilation of vascular beds mediated by shear forces within the microvasculature and metabolic activity in the muscle; and (4) lumped oxygenation of the skeletal muscle using a coupled set of equations representing oxygen transport in the blood and muscle fibres and cellular metabolism [8]. Ultrasound and NIRS data during transition to exercise will be used to validate the numerical model. By linking vascular anatomy and function, plaque severity, and muscle metabolism on oxygen dynamics, this work aims to help identify key causes of IC in PAD patients and guide targeted strategies for improved diagnosis and treatment.

References

[1] Houghton, J. S. M., Saratzis, A. N., Sayers, R. D., & Haunton, V. J. (2024). New Horizons in Peripheral Artery Disease. Age and Ageing, 53(6), afae114.

[2] Fowkes, F. G. R., Rudan, D., Rudan, I., Aboyans, V., Denenberg, J. O., McDermott, M. M., Norman, P. E., Sampson, U. K. A., Williams, L. J., Mensah, G. A., & Criqui, M. H., 2013, Comparison of global estimates of prevalence and risk factors for peripheral artery disease in 2000 and 2010: a systematic review and analysis. The Lancet, 382(9901), pp. 1329-1340.

[3] Sprenger, L., Mader, A., Larcher, B., Mächler, M., Vonbank, A., Zanolin-Purin, D., Leiherer, A., Muendlein, A., Drexel, H., & Saely, C. H. (2021). Type 2 diabetes and the risk of cardiovascular events in peripheral artery disease versus coronary artery disease. BMJ Open Diabetes Research & Care, 9(2), e002407.

[4] Parwani, D., Ahmed, M. A., Mahawar, A., & Gorantla, V. R. (2023). Peripheral Arterial Disease: A Narrative Review. Cureus, 15(6), e40267.

[5] Pabon, M., Cheng, S., Altin, S. E., Sethi, S., Nelson, M. D., et al. (2022). Sex differences in peripheral artery disease. Circulation Research, 130(4), 496-511.

[6] Alastruey, J., Parker, K. H., & Sherwin, S. J. (2012). Arterial pulse wave haemodynamics. In S. Anderson (Ed.), 11th International Conference on Pressure Surges (pp. 401-443). Virtual PiE Led t/a BHR Group.

[7] Karshafian, R., Burns, P. N., & Henkelman, M. R. (2003). Transit time kinetics in ordered and disordered vascular trees. Physics in medicine and biology, 48(19), 3225-3237.

[8] Lai, N., Camesasca, M., Saidel, G.M. et al. (2007) Linking Pulmonary Oxygen Uptake, Muscle Oxygen Utilization and Cellular Metabolism during Exercise. Ann Biomed Eng 35, 956-969

Shape optimization using PhiFEM

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Submission type: Poster

Presenter: Raphaël Bulle (Inria de l'Université de Lorraine)

The level-set method for evolving interfaces was first introduced in [1] and has been applied to a variety of shape optimization problems (see e.g. [3,4,5,6,7,8]). The core idea of this method is to use a level-set description of the domain to be optimized and modify this level-set e.g. by the derivation of optimality conditions via the shape derivative of the domain [3].

Like the level-set method, the φ-FEM takes advantage of a description of the domain via a level-set. The φ-FEM is an immersed boundary finite element method (FEM) with optimal convergence rate and conditioning which does not require any non-standard quadrature rule on cut cells nor non-standard finite element shape functions [9,10]. Thus, the φ-FEM is naturally well suited to be used with the level-set method to perform shape optimization as it provides a good description of the evolving boundary without any re-meshing of the domain.

In this talk, we describe a preliminary study on a shape optimization algorithm using φ-FEM to solve the state equations. This algorithm is implemented in FEniCSx via the recent FormOpt shape optimization toolbox [8] and the phiFEM python package [11]. We investigate the benefits of using φ-FEM instead of a standard non-conforming FEM by performing several numerical comparisons on different types of shape optimization problems.

References

[1] Osher, Stanley, and James A. Sethian. 1988. "Fronts Propagating with Curvature-Dependent Speed: Algorithms Based on Hamilton-Jacobi Formulations." Journal of Computational Physics 79(1):12-49. DOI: 10.1016/0021-9991(88)90002-2.

[2] Sethian, J. A. 1999. Level Set Methods and Fast Marching Methods: Evolving Interfaces in Computational Geometry, Fluid Mechanics, Computer Vision, and Materials Science. Cambridge University Press.

[3] Allaire, Grégoire, François Jouve, and Anca-Maria Toader. 2002. "A Level-Set Method for Shape Optimization." Comptes Rendus Mathematique 334(12):1125-30. DOI: 10.1016/S1631-073X(02)02412-3.

[4] Feppon, F., G. Allaire, F. Bordeu, J. Cortial, and C. Dapogny. 2019. "Shape Optimization of a Coupled Thermal Fluid-Structure Problem in a Level Set Mesh Evolution Framework." SeMA Journal 76(3):413-58. DOI: 10.1007/s40324-018-00185-4.

[5] Maury, Aymeric, Grégoire Allaire, and François Jouve. 2018. "Elasto-Plastic Shape Optimization Using the Level Set Method." SIAM Journal on Control and Optimization 56(1):556-81. DOI: 10.1137/17M1128940.

[6] Maury, Aymeric, Grégoire Allaire, and François Jouve. 2017. "Shape Optimisation with the Level Set Method for Contact Problems in Linearised Elasticity." The SMAI Journal of Computational Mathematics 3:249-92. DOI: 10.5802/smai-jcm.27.

[7] Laurain, Antoine. 2018. "A Level Set-Based Structural Optimization Code Using FEniCS." Structural and Multidisciplinary Optimization 58(3):1311-34. DOI: 10.1007/s00158-018-1950-2.

[8] Díaz-Avalos, Josué D., and Antoine Laurain. 2026. "FormOpt: A FEniCSx Toolbox for Level Set-Based Shape Optimization Supporting Parallel Computing." DOI: 10.48550/arXiv.2601.05709.

[9] Duprez, Michel, and Alexei Lozinski. 2020. "φ-FEM: A Finite Element Method on Domains Defined by Level-Sets." SIAM Journal on Numerical Analysis 58(2):1008-28. DOI: 10.1137/19M1248947.

[10] Duprez, Michel, Vanessa Lleras, and Alexei Lozinski. 2023. "A New φ-FEM Approach for Problems with Natural Boundary Conditions." Numerical Methods for Partial Differential Equations 39(1):281-303. DOI: 10.1002/num.22878.

[11] Bulle, Raphael, Michel Duprez, and Killian Vuillemot. 2025. "phiFEM: A Convenience Package for Using phiFEM with FEniCSx." DOI: 10.6084/M9.FIGSHARE.30373546.

IMPACT - Modelling and optimisation of endless composite filament wound armour plates

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Submission type: Poster

Presenter: Paul T. Kühner (University of Luxembourg)

Sparse, wound carbon-reinforced structures are interesting for their high stiffness to weight ratio. Especially in the design of lightweight vehicle armour plates. However, the plates are often subject to a ballistic impact, which brings the structure close to (or beyond) its operational limits. This is contributed to by the onset of failure mechanisms such as buckling, fracture, shock-waves or other dynamic effects. We present a numerical model and optimisation of these structures. The implementation of the model developed during this project is open source and builds on top of `FEniCSx`. In particular employing non-standard functionality for the parallel ready implementation, including vertex integrals and branching manifolds. A novel geometry description of geometrically-discontinuous truss-like structures is used [1]. This description is in particular useful in the optimisation, due to a smaller number of intrinsic degrees-of-freedom.

References

1. Kühner, P. T., Habera, M., Zilian, A. (18 March 2026). Modelling of sparse lightweight composite structures using a geometrically discontinuous approach. GAMM 2026, Stuttgart, Germany.

What's new in dolfiny - high-level convenience wrappers for DOLFINx

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Submission type: Poster

Presenter: Andreas Zilian (University of Luxembourg)

Dolfiny is a collection of convenience wrappers and extended functionality for DOLFINx, first presented in FEniCS 2021 [1] and continuously developed. Over time, the number of features has grown and covers wrappers for PETSc SNES and SLEPc, static condensation, arc-length load control, ODE integration, function space restriction and other minor helpers. More recently we have added support for physical units [2], dimensional analysis [3] and a PETSc TAO wrapper [4]. In this poster contribution we provide an overview of its capabilities and plans for the future.

References

1. Zilian, A., Habera, M. (23 March 2021). dolfiny: Convenience wrappers for DOLFINx. FEniCS 2021, Cambridge, United Kingdom.

2. Habera, M., Zilian, A. (18 June 2025). Physical units in UFL. FEniCS 2025, Groningen, Netherlands.

3. Zilian, A., Habera, M. (20 June 2025). Dimensional analysis in UFL. FEniCS 2025, Groningen, Netherlands.

4. Kühner, P. T., Habera, M., Zilian, A. (18 June 2025). PETSc TAO support for optimisation problems with FEniCSx. FEniCS 2025, Groningen, Netherlands.

Adaptive edge element method for a quasilinear problem in electromagnetism with strong convergence for Gauss' law

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Submission type: Poster

Presenter: Sanyang Liu (The University of Hong Kong)

This paper is concerned with adaptive edge element methods based on a regularized formulation([1]-[3]) for a quasilinear H(curl)-elliptic problem from electromagnetism. The lowest order 1st-kind and 2nd-kind Nédélec elements are employed in discrete problems. Our theoretical results include an H(curl)-strong convergence of the sequence of discrete solutions by the proposed adaptive algorithm towards the exact solution and a vanishing limit of the sequence of relevant error estimators. Moreover, the adaptive algorithm is shown to ensure a strong convergence for the associated discrete Gauss' law in the H{-1}-norm. Finally, with the Dörfler marking included, the adaptive algorithm is shown to be contractive for the sum of an H(curl)-error related quantity and a scaled estimator up to a high order term between two consecutive adaptive loops.

References

[1] Ciarlet, P., Wu, H., & Zou, J. (2014). Edge Element Methods for Maxwell's Equations with Strong Convergence for Gauss' Laws. SIAM Journal on Numerical Analysis, 52(2), 779-807. DOI: 10.1137/120899856

[2] Duan, H., Li, S., Tan, R. C., & Zheng, W. (2012). A delta-regularization finite element method for a double curl problem with divergence-free constraint. SIAM Journal on Numerical Analysis, 50(6), 3208-3230. DOI: 10.1137/110850578

[3] Reitzinger, S. and Schöberl, J. (2002), An algebraic multigrid method for finite element discretizations with edge elements. Numer. Linear Algebra Appl., 9: 223-238. DOI: 10.1002/nla.271

Flow distortion generation using topology optimization

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Submission type: Poster

Presenter: Langlet Nathan (DAAA, ONERA, Institut Polytechnique de Paris, 92190, Meudon, France)

This study investigates the use of topology optimization to automatically identify aerodynamic designs that accurately generate a specific target velocity profile. The methodology couples the steady-state RANS equations with the Spalart-Allmaras turbulence model and a relaxed eikonal equation for dynamic wall-distance computation, implemented using the FEniCSx library. To ensure clearly defined solid-fluid interfaces, the Topology Optimization of Binary Structures (TOBS) method is employed, utilizing integer linear programming via the CPLEX software. The approach is validated on the wake reproduction of a square cylinder at Re = 105, and further applied to flow separation control in a 90° sharp-angled bend at Re = 104. Results demonstrate that the framework successfully creates streamlined guide vanes that eliminate flow separation, maintaining target velocity profiles within a $\pm 5\%$ tolerance. The influence of initial guesses and optimization parameters is analyzed, highlighting the existence of multiple local minima and the validation of the method for complex turbulent flows. The governing equations include incompressible RANS with Brinkman term for solid modeling, and the S-A model with penalty for turbulence in solids. The objective is a weighted sum of velocity error at AIP and regularization. Numerical implementation uses Newton solver for nonlinear system and branch-and-bound for ILP. In the square case, objective reduces 380-fold, with geometry matching the target wake. For the bend, four cases show curved vanes, with convergence varying by initial guess, revealing TOBS efficiency in adding material. Conclusion suggests extensions to 3D compressible flows and TOBS-GT for better interface.

Effects of Pre-existing Defects on Helium Bubble Nucleation and Growth in Tungsten

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Submission type: Poster

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The tokamak is a machine designed to produce nuclear fusion reactions, enabling the generation of abundant, sustainable, low-carbon energy. One of its essential components is the divertor, made of tungsten monoblocks. Helium, produced by fusion reactions, becomes embedded in the tungsten and causes the formation of bubbles and other microstructural defects [1]. These changes gradually alter the mechanical and thermal properties of the material. Some models describe bubble formation through a self-trapping mechanism, but there is still no model intrinsic defect (both vacancies and dislocations) in the material.

In this work, we model a cluster dynamics problem in a continuous medium describing the formation of helium bubbles by trap mutation and nucleation at dislocations and vacancies. The evolution of bubble concentrations promotes their coalescence. Based on previous model, solved with legacy FEniCS [2], we develop a one-dimensional finite element model on FEniCSx, with a variational formulation of the coupled equations governing these phenomena and a bursting model [3]. The resulting nonlinear system is assembled using the Unified Form Language (UFL) formalism and solved by the PETSc nonlinear solvers integrated into DOLFINx, via Krylov linearization to ensure numerical robustness and convergence using recommendations from [4].

The model allows us to track the spatial and temporal distributions of helium concentrations, as well as the evolution of the bubble radius. We focus on the coupling equations between three sources of nucleation (trap mutation, vacancies and dislocations) and bubbles. We introduce an additional to describe the reduction of nucleation rate by annihilation of vacancies and dislocations when bubbles are created. We show the precautions necessary in this case to achieve convergence.

References

[1] Dutta, N.J. et al., Journal of Nuclear Materials 452 (2014) 51-56.

[2] Delaporte-Mathurin, R. et al., Scientific Reports 11 (2021) 14681.

[3] Pappalardo, F. et al., Journal of Physics D: Applied Physics 59 (2026) 085203.

[4] Balay, S. et al., Argonne National Laboratory (2024).

Finite Element Modeling of Electrochemical Impedance Spectroscopy in All-Solid-State Batteries using FEniCSx

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All-solid-state thin-film batteries are key components for the development of autonomous integrated systems, where performance depends on complex charge transport and interface kinetics [1]. Electrochemical Impedance Spectroscopy (EIS) represents a non-destructive technique used to characterize these phenomena by decoupling the resistive and capacitive contributions of the solid electrolyte and its interfaces [2]. In this work, we implement a numerical framework within FEniCSx to simulate the EIS response of such systems based on the Poisson-Nernst-Planck (PNP) model. This model couples electrostatic potential and ionic concentrations, accounting for the formation of the electrical double layers at the electrode-electrolyte interfaces [3]. We utilize the Unified Form Language (UFL) to perform an automatic linearization of the non-linear coupled system around various electrode bias potentials. This approach allows for a direct resolution in the frequency domain. The central feature of the implementation is the post-processing of the frequency-domain results, using a residual-based current calculation to ensure strict electrical conservation. We demonstrate the framework's validity by comparing simulated Nyquist plots with experimental data obtained from thin -film batteries. This work provides a physically consistent tool for the study of electrochemical phenomena in solid-state energy storage.

References

[1] Wu, T., Dai, W., Ke, M., Huang, Q., & Lu, L. (2021). All-Solid-State Thin Film μ-Batteries for Microelectronics. Advanced Science, 8(19), 2100774. DOI: 10.1002/advs.202100774

[2] Orazem, M. E., & Tribollet, B. (2020). A tutorial on electrochemical impedance spectroscopy. ChemTexts, 6(2), 12. DOI: 10.1007/s40828-020-0110-7

[3] Bazant, M. Z., Thornton, K., & Ajdari, A. (2004). Diffuse-charge dynamics in electrochemical systems. Physical Review E, 70(2), 021506. DOI: 10.1103/PhysRevE.70.021506

GPU kernels in DOLFINx

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Submission type: Poster

Presenter: Chris Richardson (University of Cambridge)

There are many different ways that we can think about using GPUs for Finite Element Modelling. Over the last few years, along with others, we have been working on various different projects that use GPUs with FEniCSx. The simplest approach is to use the existing CPU code for matrix assembly, and copy the sparse matrix to the GPU. It doesn't require much (or sometimes any) code changes: we rely on the solver to handle the details. This approach can be used with PETSc/hypre, Ginkgo, amgX and CUDSS (NVIDIA's direct solver). Despite this simplicity, we are still faced with the issue of binding CPU to GPU - at the moment this is done by one CPU core to one GPU, which makes the mesh processing relatively slow. We are developing threading techniques on CPU to speed this up. The next level of complexity involves on-device assembly: first, we need to prepare a sparsity pattern on the device, then accumulate into that sparse data structure. Since ffcx-generated code is not very GPU-friendly, we write custom code, pre-computing geometric factors, which can be streamed from memory. For higher-order problems, matrices are not efficient, and we have developed some matrix-free methods for p-multigrid, which can run in parallel across a HPC system with hundreds of GPU devices to solve very large problems. There is also some work on Python wrappers for GPU data, that allow interoperability with `cupy` arrays, which works for both NVIDIA and AMD GPUs.

References

DOI: 10.1016/j.parco.2023.103051, https://scientificcomputing.github.io/fenics2024/gpu-acceleration, DOI: 10.1016/j.procs.2025.08.235

Numerical simulation of asphalt solar collector systems

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Submission type: Poster

Presenter: Lucia Escudero Sartages (Centre de Recerca Matematica, Spain)

Urban pavements absorb large amounts of solar radiation, raising surface temperatures and intensifying the urban heat island effect. These conditions reduce urban comfort, accelerate pavement deterioration, and increase cooling energy demand in surrounding buildings. To address these challenges, previous research has explored technologies to harness stored pavement heat, including geothermal paving systems and hybrid photovoltaic-thermal pavements. Nevertheless, challenges remain related to long-term pavement durability, climatic variability, and the optimisation of collector geometry and material properties. The ENHANCE Europe project investigates an innovative solution based on asphalt solar collectors embedded within flexible pavement structures. The system aims to recover excess heat stored in the pavement and convert it into a usable thermal resource for nearby buildings, reducing fossil-fuel consumption, lowering operational energy costs, and enhancing the resilience of urban energy systems. Within this framework, the present work introduces a finite-element numerical model for simulating coupled heat transfer and fluid flow in multilayer pavements with embedded collector networks. The model solves the incompressible Navier-Stokes and transient heat equations using an Incremental Pressure Correction Scheme. The simulations are implemented in DOLFINx, while the computational domain and mesh are generated using Gmsh. The model provides temperature distributions at critical pavement depths, quantifies heat extraction efficiency, and evaluates system performance under varying environmental and operational conditions. The numerical results support informed design and construction decisions, enabling the optimisation of system geometry, material selection, and operating strategies. Future work will focus on model refinement using experimental data from four European living labs, contributing to digital and data-driven tools that support the urban energy transition.

References

Bobes-Jesús, V., Pascual-Muñoz, P., Castro-Fresno, D., & Rodríguez-Hernández, J. (2013). Asphalt solar collectors: A literature review. Applied Energy, 102, 962-970. DOI: 10.1016/j.apenergy.2012.08.050

Qin, Y., & Hiller, J. E. (2014). Understanding pavement-surface energy balance and its implications on cool pavement development. Energy and Buildings, 85, 389-399. DOI: 10.1016/j.enbuild.2014.09.076

Dokken, J. S., Johansson, A., Massing, A. & Funke, S.W. (2020). A multimesh finite element method for the Navier-Stokes equations based on projection methods. Computer Methods in Applied Mechanics and Engineering, 368, 113129. DOI: 10.1016/j.cma.2020.113129

Integration of Externally Defined Constitutive Models into FEniCSx Using DOLFINx-External-Operator

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Submission type: Poster

Presenter: Andrey Latyshev (University of Luxembourg, Institut Jean Le Rond d'Alembert, Sorbonne Université)

The Unified Form Language (UFL) (Alnaes et al., 2013) is essential for writing variational forms of partial differential equations (PDEs) in automated finite element solvers like FEniCS. However, UFL has a critical limitation: non-trivial solid mechanics problems, such as complex plasticity or neural network constitutive models, cannot be naturally formulated within it. This restriction has slowed FEniCS adoption in the broader solid mechanics community.

To overcome this bottleneck, we introduce DOLFINx-External-Operator (Latyshev et al., 2025), a novel software framework for FEniCSx. Based on DOLFINx's new data-centric design (Baratta et al., 2023), automatic UFL Expression code generation, and the UFL ExternalOperator extension (Bouziani & Ham, 2021), this tool enables the seamless integration of arbitrary constitutive models via third-party packages that support ndarray-like data structures, such as JAX (Bradbury et al., 2018) and Numba (Lam et al., 2015).

As a particular outcome, we leverage JAX for forward-mode algorithmic automatic differentiation (AD), eliminating the need for error-prone manual derivatives. We demonstrate the use of AD on a complex non-associative Mohr-Coulomb elastoplastic model with apex smoothing (Helfer et al., 2024). The framework and examples are open-source under the LGPLv3 license.

References

Alnæs, M. S., Logg, A., Ølgaard, K. B., Rognes, M. E., & Wells, G. N. (2014). Unified form language: A domain-specific language for weak formulations of partial differential equations. ACM Transactions on Mathematical Software, 40(2), 9:1-9:37. DOI: 10.1145/2566630

Baratta, I. A., Dean, J. P., Dokken, J. S., Habera, M., Hale, J. S., Richardson, C. N., Rognes, M. E., Scroggs, M. W., Sime, N., & Wells, G. N. (2023). DOLFINx: The next generation FEniCS problem solving environment. Zenodo. DOI: 10.5281/zenodo.10447666

Bouziani, N., & Ham, D. A. (2021). Escaping the abstraction: A foreign function interface for the Unified Form Language [UFL] (arXiv:2111.00945). arXiv. DOI: 10.48550/arXiv.2111.00945

Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Katariya, Y., Leary, C., Maclaurin, D., Necula, G., Paszke, A., Vander-Plas, J., Wanderman-Milne, S., & Zhang, Q. (2018). JAX: Composable transformations of Python+NumPy programs (Version 0.3.13) [Computer software]. http://github.com/jax-ml/jax

Helfer, T., Michel, B., Proix, J.-M., Sercombe, J., Casella, M., & Salvo, M. (2024). Invariant-based implementation of the Mohr-Coulomb elasto-plastic model in OpenGeoSys using MFront (Version 4.2.1) [Computer software]. https://thelfer.github.io/tfel/web/MohrCoulomb.html

Latyshev, A., Bleyer, J., Maurini, C., & Hale, J. (2025). Expressing general constitutive models in FEniCSx using external operators and algorithmic automatic differentiation. Journal of Theoretical, Computational and Applied Mechanics. DOI: 10.46298/jtcam.14449

Towards natural language-driven computational mechanics: A general LLM-integrated platform built on FEniCS

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Hong Kong University of Science and Technology

Submission type: Software Demonstration

Presenter: Guangjin Mou (Postdoctoral researcher in HKUST)

The increasing complexity of computational mechanics workflows, particularly in finite element analysis, creates a significant gap between high-level modeling intent and low-level numerical implementation. Recent advances in large language models (LLMs) offer a promising pathway to bridge this gap by enabling natural language interaction with scientific computing systems.

In this work, we present a general language-driven computational mechanics platform that integrates LLMs with the FEniCS ecosystem for PDE-based modeling and simulation. Building upon a modular agent architecture, the proposed framework connects user queries, LLM reasoning, and physics-based solvers through a structured tool interface. The system leverages a Model Context Protocol (MCP)-inspired design to expose FEniCS functionalities such as variational form definition, boundary condition specification, and solver execution as callable tools, enabling robust and controllable interaction between the LLM and the numerical backend.

Unlike existing approaches that are tightly coupled to specific numerical frameworks, the proposed platform exploits the symbolic variational formulation of FEniCS to support a wide range of problems in computational mechanics, including heat conduction, linear and nonlinear elasticity, and multi-physics simulations. The LLM operates as an orchestration layer that interprets user intent, constructs problem definitions, selects appropriate tools, and manages execution workflows, while explicit instruction design ensures reliability and prevents uncontrolled behavior.

The proposed framework transforms traditional finite element workflows into an interactive, language-driven process, significantly lowering the barrier to complex PDE modeling. This work demonstrates the feasibility of a new paradigm in computational mechanics, where natural language serves as a high-level interface for general-purpose simulation and design.

CutFEMx: a cut finite element library for FEniCSx

Susanne Claus

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Submission type: Software Demonstration

Presenter: Susanne Claus (ONERA)

CutFEMx is an extension to FEniCSx for cut finite element methods, released under the MIT license and available at https://github.com/sclaus2/CutFEMx. It extends DOLFINx with support for unfitted discretisations based on level set geometries, enabling the solution of partial differential equations on complex and evolving domains without meshing.

CutFEMx builds on two additional open-source components. The first, CutCells (MIT, available at https://github.com/sclaus2/CutCells), provides robust and efficient algorithms for computing intersections between level set functions and finite element cells of various types. It supports higher-order cutting up to polynomial degree (p=4) and generates both cut meshes for visualisation and quadrature rules on cut cells. The second component, runintgen (MIT, available at https://github.com/sclaus2/runintgen), extends FFCx with runtime quadrature kernel generation, allowing the integration of UFL forms over quadrature rules that vary between elements and are only defined at runtime, as required for cut cell integration.

The framework is designed with an emphasis on performance and modularity, combining cell-local operations, zero-copy data exchange, and compatibility with automatic differentiation workflows. The capabilities of CutFEMx are illustrated on representative examples, including Poisson, Stokes, and elasticity problems posed on implicitly defined domains. Application areas include shape and topology optimization, fluid-structure interaction, multi-material problems, and simulations involving evolving interfaces, where unfitted finite element methods provide a flexible alternative to mesh-conforming approaches.

Software demonstration (GPU)

Chris Richardson, Joe Dean, Garth Wells
University of Cambridge

Submission type: Software Demonstration

Presenter: Chris Richardson (University of Cambridge)

Some online demonstration of GPU code, see through the terminal. Just a chance to try out some things which are on the poster.

Numerical investigation of urban heat island within variable porous urban domain

Luis Gerardo Gutierrez Ibarra

University of Guadalajara

Submission type: Software Demonstration

Presenter: Luis Gerardo Gutierrez Ibarra (University of Guadalajara)

This study presents a numerical analysis of the Urban Heat Island (UHI) effect, defined as the temperature differential between urban and rural areas. To efficiently mesh and simulate large-scale cities, urban morphology is modeled as a variable porous medium, where buildings and streets represent the solid and fluid phases, respectively. We analyze an 11000 mts x 800 mts domain with a concentric porosity distribution-dense at the city center and sparse in the suburbs-driven by real meteorological solar radiation data. The computational solver is built on FEniCSx, extending Jørgen S. Dokken's fractional-step Navier-Stokes pipeline to include porous media resistance and thermal coupling. For the spatial discretization, Taylor-Hood elements are used to ensure velocity-pressure stability, augmented by SUPG (Streamline-Upwind/Petrov-Galerkin) and GLS (Galerkin/Least-Squares) stabilizers. Temporal discretization is handled via a second-order Backward Differentiation Formula (BDF2) integrated into an Incremental Pressure Correction Scheme (IPCS), a standard variation of the Chorin projection method. We present interesting results from the wind velocity field, demonstrating that the variable porous medium induces von Kármán vortices. This phenomenon occurs despite the simulation not yet resolving a full turbulence model (a feature currently under development). This effect is analyzed across different city morphologies (dense vs. dispersed) to visualize their respective impacts on air temperature and overall UHI intensity.

Live Demonstration of Z3ST: A FEniCSx Framework for Multiscale Thermo-Mechanical Analysis with Automatic Differentiation

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Submission type: Software Demonstration

Presenter: Giovanni Zullo (Politecnico di Milano)

This software demonstration showcases Z3ST, an open-source framework for multiphysics thermo-mechanical simulations with automatic differentiation, built on FEniCSx (Baratta et al., 2023; Scroggs et al., 2022). The demonstration focuses on crystal plasticity and automated constitutive law discovery, highlighting how UFL symbolic differentiation (Alnaes et al., 2014; Latyshev et al., 2025) enables rapid development of complex material models. The demonstration walks through a single-crystal viscoplastic deformation simulation under uniaxial loading, showing how automatic differentiation eliminates manual Jacobian derivation with good convergence properties. The implementation includes FCC slip systems with power-law kinetics and history variables managed through quadrature function spaces. A second highlight is automated discovery of constitutive laws (Flaschel et al., 2022). The demonstration shows how gradients computed via automatic differentiation enable training interpretable material models from data, bridging machine learning and computational mechanics for irradiated materials (Frydrych, 2023). The practical workflow is demonstrated: YAML-based simulation setup, custom constitutive laws defined as Python functions with automatic differentiation, convergence monitoring, and visualization using PyVista. Pre-computed results illustrate broader capabilities including thermo-mechanical design-by-analysis for pressure vessels and microstructural analysis of over-pressurized bubbles with phase-field fracture. All code is open-source with 30+ verification cases. Code availability: https://github.com/giozu/z3st.

References

Alnaes, M. S., Logg, A., Olgaard, K. B., Rognes, M. E., & Wells, G. N. (2014). Unified form language: A domain-specific language for weak formulations of partial differential equations. ACM Transactions on Mathematical Software, 40(2). DOI: 10.1145/2566630

Baratta, I. A., Dean, J. P., Dokken, J. S., Habera, M., Hale, J. S., Richardson, C. N., Rognes, M. E., Scroggs, M. W., Sime, N., & Wells, G. N. (2023). DOLFINx: The next generation FEniCS problem solving environment. Zenodo. DOI: 10.5281/zenodo.10447666

Flaschel, M., Kumar, S., & De Lorenzis, L. (2022). Discovering plasticity models without stress data. npj Computational Materials, 8, 91. DOI: 10.1038/s41524-022-00752-4

Frydrych, K. (2023). Modelling irradiation effects in metallic materials using the crystal plasticity theory - A review. Crystals, 13(5), 771. DOI: 10.3390/cryst13050771

Latyshev, A., et al. (2025). Expressing general constitutive models in FEniCSx using external operators and algorithmic automatic differentiation. Journal of Theoretical, Computational and Applied Mechanics. DOI: 10.46298/jtcam.14449

Scroggs, M. W., Dokken, J. S., Richardson, C. N., & Wells, G. N. (2022). Construction of arbitrary order finite element degree-of-freedom maps on polygonal and polyhedral cell meshes. ACM Transactions on Mathematical Software, 48(2), Article 18. DOI: 10.1145/3524456

Live Demonstration of an Open-Source Vibroacoustic GUI Built on FEniCSx

Antonio Baiano Svizzero

Undabit

Submission type: Software Demonstration

Presenter: Antonio Baiano Svizzero (Undabit)

We demonstrate a fully open-source desktop application for vibroacoustic simulation built on top of FEniCSx. The software provides a graphical user interface, developed with PyQt6 and PyVista, that guides the user through the complete simulation workflow: importing a finite element mesh (including models originally created in Nastran format), defining materials and boundary conditions, assembling and solving the coupled fluid-structure problem, and visualising the results interactively.

FEniCSx in Metal Additive Manufacturing

Anton Evdokimov

heatshape GmbH

Submission type: Software Demonstration

Presenter: Anton Evdokimov (heatshape GmbH)

heatshape is a cloud-based simulation software for welding, metal additive manufacturing, and heat treatment. It combines multiple open-source software solutions to enable user-friendly model setup and efficient computation. This demonstration will present the software itself as well as its internal architecture and discuss how different open-source components are integrated into a unified simulation platform.

From Micro to Macro: Unsupervised FE2 Acceleration using ddfenicsx and micmacsfenicsx

Felipe Rocha

Université Paris-Est Créteil - MSME

Submission type: Software Demonstration

Presenter: Felipe Rocha (Université Paris-Est Créteil - MSME)

The FE2 approach [4] consists of solving a multilevel nonlinear finite element problem and is typically applied to problems with clear scale separation, where microscopic heterogeneities are much smaller than the structural scale. Resolving these heterogeneities directly at the structural level would require prohibitively fine meshes. Computational homogenisation provides consistent coupling between scales [5] (e.g. coarse-scale stress is the average of the microscale stress field). However, FE2 remains computationally expensive and memory intensive, which limits its practical use. This tutorial presents the synergic use of two libraries: i) ddfenicsx [1], which integrates Data-Driven Computational Mechanics (DDCM) [6] while preserving a familiar UFL-based workflow (see the ddfenicsx presentation abstract for details); and ii) micmacsfenics [2], an FE2 implementation featuring (i) nested FEM solvers for local problems at each coarse-scale Gauss point, (ii) efficient computation of consistent homogenised tangents (no finite differences), and (iii) support for common non-local microscale kinematical constraints (e.g. periodic, zero-average, etc). The proposed acceleration relies on a recent approach [3] that constructs strain-stress datasets on-the-fly through computational homogenisation. The method is fully unsupervised: no offline database generation or neural-network training is required [7]. Although the method focuses on solid mechanics, the methodology and libraries are general and can be easily extended to other applications. Attendees will run: i) a single-scale hyperelastic problem; ii) a heterogeneous microscale model using micmacsfenics; iii) a reference FE2 solution; and iv) a DDCM problem with a self-evolving dataset generated by the microscale solver using ddfenicsx. All scripts will be provided in a GitHub repository. Requirements are FEniCSx 0.10, dolfin_mpc, and scikit-learn.

References

[1] Rocha, F. (2023-). ddfenics: A FEniCS(x)-based (model-free) data-driven computational mechanics implementation [Software]. Zenodo. DOI: 10.5281/zenodo.7646226

[2] Rocha, F. (2023-). micmacsfenics: A FEniCs-based implementation of two-level finite element simulations (FE2) using computational homogenization [Software]. Zenodo. DOI: 10.5281/zenodo.7643777

[3] Rocha, F., Platzer, A., Leygue, A., & Stainier, L. (2025). On-the-fly adaptive sampling strategy for data-driven computational mechanics: Applications to computational homogenisation. Mechanics of Materials, 207, 105382. DOI: 10.1016/j.mechmat.2025.105382

[4] Feyel, F. (2003). A multilevel finite element method (FE2) to describe the response of highly non-linear structures using generalized continua. Computer Methods in Applied Mechanics and Engineering, 192(28-30), 3233-3244. DOI: 10.1016/S0045-7825(03)00348-7

[5] de Souza Neto, E. A., Blanco, P. J., Sánchez, P. J., & Feijóo, R. A. (2015). An RVE-based multiscale theory of solids with micro-scale inertia and body force effects. Mechanics of Materials, 80(A), 136-144. DOI: 10.1016/j.mechmat.2014.10.007

[6] Kirchdoerfer, T., & Ortiz, M. (2016). Data-driven computational mechanics. Computer Methods in Applied Mechanics and Engineering, 304, 81-101. DOI: 10.1016/j.cma.2016.02.001

[7] Zlatic, M., Rocha, F., Stainier, L., & Canadija, M. (2024). Data-driven methods for computational mechanics: A fair comparison between neural networks based and model-free approaches. Computer Methods in Applied Mechanics and Engineering, 431, 117289. DOI: 10.1016/j.cma.2024.117289

Integration of Externally Defined Constitutive Models into FEniCSx Using DOLFINx-External-Operator

Andrey Latyshev, Jørgen S. Dokken, Jérémy Bleyer, Corrado Maurini, Jack S. Hale

University of Luxembourg, Institut Jean Le Rond d'Alembert, Sorbonne Université, Simula Research Laboratory, Laboratoire Navier, École des Ponts ParisTech, Université Gustave Eiffel

Submission type: Software Demonstration

Presenter: Andrey Latyshev (University of Luxembourg, Institut Jean Le Rond d'Alembert, Sorbonne Université)

The Unified Form Language (UFL) (Alnaes et al., 2013) is essential for writing variational forms of partial differential equations (PDEs) in automated finite element solvers like FEniCS. However, UFL has a critical limitation: non-trivial solid mechanics problems, such as complex plasticity or neural network constitutive models, cannot be naturally formulated within it. This restriction has slowed FEniCS adoption in the broader solid mechanics community.

To overcome this bottleneck, we introduce DOLFINx-External-Operator (Latyshev et al., 2025), a novel software framework for FEniCSx. Based on DOLFINx's new data-centric design (Baratta et al., 2023), automatic UFL Expression code generation, and the UFL ExternalOperator extension (Bouziani & Ham, 2021), this tool enables the seamless integration of arbitrary constitutive models via third-party packages that support ndarray-like data structures, such as JAX (Bradbury et al., 2018) and Numba (Lam et al., 2015).

As a particular outcome, we leverage JAX for forward-mode algorithmic automatic differentiation (AD), eliminating the need for error-prone manual derivatives. We demonstrate the use of AD on a complex non-associative Mohr-Coulomb elastoplastic model with apex smoothing (Helfer et al., 2024). The framework and examples are open-source under the LGPLv3 license.

References

Alnæs, M. S., Logg, A., Ølgaard, K. B., Rognes, M. E., & Wells, G. N. (2014). Unified form language: A domain-specific language for weak formulations of partial differential equations. ACM Transactions on Mathematical Software, 40(2), 9:1-9:37. DOI: 10.1145/2566630

Baratta, I. A., Dean, J. P., Dokken, J. S., Habera, M., Hale, J. S., Richardson, C. N., Rognes, M. E., Scroggs, M. W., Sime, N., & Wells, G. N. (2023). DOLFINx: The next generation FEniCS problem solving environment. Zenodo. DOI: 10.5281/zenodo.10447666

Bouziani, N., & Ham, D. A. (2021). Escaping the abstraction: A foreign function interface for the Unified Form Language [UFL] (arXiv:2111.00945). arXiv. DOI: 10.48550/arXiv.2111.00945

Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Katariya, Y., Leary, C., Maclaurin, D., Necula, G., Paszke, A., Vander-Plas, J., Wanderman-Milne, S., & Zhang, Q. (2018). JAX: Composable transformations of Python+NumPy programs (Version 0.3.13) [Computer software]. http://github.com/jax-ml/jax

Helfer, T., Michel, B., Proix, J.-M., Sercombe, J., Casella, M., & Salvo, M. (2024). Invariant-based implementation of the Mohr-Coulomb elasto-plastic model in OpenGeoSys using MFront (Version 4.2.1) [Computer software]. https://thelfer.github.io/tfel/web/MohrCoulomb.html

Latyshev, A., Bleyer, J., Maurini, C., & Hale, J. (2025). Expressing general constitutive models in FEniCSx using external operators and algorithmic automatic differentiation. Journal of Theoretical, Computational and Applied Mechanics. DOI: 10.46298/jtcam.14449

FELiCS: A Versatile Linearized Flow Solver for Multi-Physics Applications

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Technische Universität Berlin

Submission type: Presentation

Presenter: Marina Matthaiou (Technische Universität Berlin)

We present FELiCS (http://felics.eu), an open-source Python package for the linearized analysis of multi-physics flows within the FEniCS ecosystem.

Understanding how small perturbations evolve in a flow is central to many engineering problems, including transition to turbulence, noise generation, and flow control. Linear analysis provides a tractable framework to study amplification mechanisms, coherent structures, and input-output behavior, even in complex and turbulent configurations.

FELiCS leverages the automated finite element formulation provided by FEniCS (Alnæs et al., 2015) to construct and discretize linearized operators derived from the incompressible and compressible Navier-Stokes equations. The framework extends naturally to heat and mass transport, turbulence modeling, and reacting flows, enabling a unified treatment of multi-physics systems.

The formulation is based on variational forms, allowing direct use of FEniCS abstractions for function spaces, boundary conditions, and unstructured meshes. The resulting linear systems and eigenvalue problems are solved using scalable solvers from PETSc (Balay et al., 2019) and SLEPc (Hernandez et al., 2005), enabling application to large-scale problems.

FELiCS adopts a modular design in which physical models are implemented as composable variational contributions. This facilitates extension to additional equations and coupled systems while maintaining consistency with the finite element formulation. The software can be used as a standalone tool via configuration files or as a Python library, enabling integration into workflows such as sensitivity analysis, shape optimization, data assimilation, and flow control.

FELiCS has been applied to a range of problems, including turbulent reacting flows and aeroacoustics (e.g., Casel et al., 2022; Demange et al., 2024). A detailed software description has been submitted to the Journal of Open Source Software.

References

Alnæs, M. S., Blechta, J., Hake, J., Johansson, A., Kehlet, B., Logg, A., Richardson, C., Ring, J., Rognes, M. E., & Wells, G. N. (2015). The FEniCS project version 1.5. DOI: 10.11588/ans.2015.100.20553

Balay, S., Abhyankar, S., Adams, M., Brown, J., Brune, P., Buschelman, K., Dalcin, L., Dener, A., Eijkhout, V., Gropp, W., et al. (2019). PETSc users manual. DOI: 10.2172/2998643

Casel, M., Oberleithner, K., Zhang, F., Zirwes, T., Bockhorn, H., Trimis, D., & Kaiser, T. L. (2022). Resolvent-based modelling of coherent structures in a turbulent jet flame using a passive flame approach. Combustion and Flame, 236, 111695. DOI: 10.1016/j.combustflame.2021.111695

Demange, S., Yuan, Z., Jekosch, S., Hanifi, A., Cavalieri, A. V. G., Sarradj, E., Kaiser, T. L., & Oberleithner, K. (2024). Resolvent model for aeroacoustics of trailing edge noise. Theoretical and Computational Fluid Dynamics. DOI: 10.1007/s00162-024-00688-z

Hernandez, V., Roman, J. E., & Vidal, V. (2005). SLEPc: A scalable and flexible toolkit for the solution of eigenvalue problems. ACM Transactions on Mathematical Software, 31(3), 351-362. DOI: 10.1145/1089014.1089019

Finite Element Modeling of Electrochemical Impedance Spectroscopy in All-Solid-State Batteries using FEniCSx

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Submission type: Presentation

Presenter: Théo Bermond (Univ. Grenoble Alpes, CEA, Leti, F-38000 Grenoble, France)

All-solid-state thin-film batteries are key components for the development of autonomous integrated systems, where performance depends on complex charge transport and interface kinetics [1]. Electrochemical Impedance Spectroscopy (EIS) represents a non-destructive technique used to characterize these phenomena by decoupling the resistive and capacitive contributions of the solid electrolyte and its interfaces [2]. In this work, we implement a numerical framework within FEniCSx to simulate the EIS response of such systems based on the Poisson-Nernst-Planck (PNP) model. This model couples electrostatic potential and ionic concentrations, accounting for the formation of the electrical double layers at the electrode-electrolyte interfaces [3]. We utilize the Unified Form Language (UFL) to perform an automatic linearization of the non-linear coupled system around various electrode bias potentials. This approach allows for a direct resolution in the frequency domain. The central feature of the implementation is the post-processing of the frequency-domain results, using a residual-based current calculation to ensure strict electrical conservation. We demonstrate the framework's validity by comparing simulated Nyquist plots with experimental data obtained from thin -film batteries. This work provides a physically consistent tool for the study of electrochemical phenomena in solid-state energy storage.

References

[1] Wu, T., Dai, W., Ke, M., Huang, Q., & Lu, L. (2021). All-Solid-State Thin Film μ-Batteries for Microelectronics. Advanced Science, 8(19), 2100774. DOI: 10.1002/advs.202100774

[2] Orazem, M. E., & Tribollet, B. (2020). A tutorial on electrochemical impedance spectroscopy. ChemTexts, 6(2), 12. DOI: 10.1007/s40828-020-0110-7

[3] Bazant, M. Z., Thornton, K., & Ajdari, A. (2004). Diffuse-charge dynamics in electrochemical systems. Physical Review E, 70(2), 021506. DOI: 10.1103/PhysRevE.70.021506

Multi-physics Modelling of Nuclear Fusion Reactor Components using FESTIM

James Dark[1], Remi Delaporte-Mathurin[1], Jorgen Dokken[2], Huihua Yang[1], Chirag Khurana[1], Kaelyn Dunnel[1]

[1] MIT
[2] Simula Research Laboratory

Submission type: Presentation

Presenter: James Dark (MIT)

The design of tritium-facing components in nuclear fusion reactors demands accurate multi-physics simulations that capture the complex interplay of physical phenomena across diverse spatial and temporal scales. This work presents a modelling approach centred on tritium transport, leveraging FESTIM, a flexible, open-source simulation tool for hydrogen isotope migration in fusion-relevant materials and geometries.

FESTIM has been developed with extensibility and interoperability in mind, making it well-suited to receive input from a range of high-fidelity physics codes. We present workflows in which outputs from OpenMC (neutron-induced tritium generation and nuclear heating) and OpenFOAM (temperature and velocity fields) are coupled with FESTIM to enable detailed, multiphysics modelling of tritium permeation, retention, and inventory under reactor-relevant conditions.

Applications include tritium migration in breeding blankets, plasma-facing components, and coupled liquid-solid domains such as heat exchangers or tritium extraction systems. In each case, external tools provide FESTIM with local source terms and operational conditions, enabling accurate evaluation of tritium transport. Validation of these multi-physics workflows has begun through collaborative projects, such as LIBRA (MIT), but further experimental validation remains a critical need for modelling tritium-facing components.

This framework supports design iteration, analysis of experimental setups, and extrapolation to reactor scale, accelerating the development of fuel-cycle components. The Python-based architecture of FESTIM also facilitates automated, scriptable workflows, making it well-suited for integration into iterative component optimisation loops. With permissive licensing and a growing user community, FESTIM is well-positioned as a flexible code for integrated multi-physics tritium transport modelling.

FESTIM v2.0: Upgraded framework for multi-species hydrogen transport

Remi Delaporte-Mathurin[1], James Dark[1], Jørgen S. Dokken[2], Kaelyn Dunnell[1], Chirag Khurana[1], Huihua Yang[1]

[1] MIT PSFC
[2] Simula

Submission type: Presentation

Presenter: Remi Delaporte-Mathurin (Massachusetts Institute of Technology, Plasma Science and Fusion Center)

FESTIM is an open-source finite element code for modelling hydrogen isotope transport in materials and fluids, developed for applications in nuclear fusion technology. Built on the FEniCS/FEniCSx framework, FESTIM enables the simulation of coupled diffusion-reaction processes including trapping, surface recombination, permeation, and multi-material transport, which are central to predicting tritium inventories and losses in fusion systems.

While FEniCS provides a powerful and flexible environment for finite element modelling, developing application-specific models from scratch can be demanding for users who are not specialists in the finite element method. FESTIM addresses this challenge by providing a high-level, domain-specific abstraction layer that allows users to define complex tritium transport problems through a simplified, physics-oriented interface, while retaining the robustness and performance of the underlying FEniCS ecosystem.

This presentation will provide an overview of the FESTIM project and its role in supporting the design and safety assessment of fusion fuel cycle components. We will illustrate how the FEniCS-based implementation enables flexible multiphysics modelling and scalable simulations across a wide range of geometries and material configurations. Application examples will be presented for key elements of the fusion fuel cycle, including breeding blankets, plasma-facing components (first wall), heat exchangers, tritium extraction systems, and isotope separation units.

A short live demonstration will highlight the user workflow, from problem definition to post-processing, and showcase recent developments based on FEniCSx. The talk will also discuss current challenges, performance considerations, and opportunities for further FEniCS community contributions.

References

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Numerical quality factor statistics of multilayered superconducting radio-frequency cavity

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Submission type: Presentation

Presenter: Aaron Gobeyn (Technical University Darmstadt)

Bulk niobium (Nb) is currently the standard material for superconducting radio-frequency (SRF) cavities for particle accelerator applications. It has been observed that the benefits of using a superconducting material occur in only a thin layer [1]. This makes a proposal made by Gurevich [2] attractive for investigation, namely, coating the SRF cavity with alternating superconducting and insulating layers, referred to as superconductor-insulator-superconductor (SIS) structures. The thin coating shields the bulk interior from accelerating fields, allowing for higher operating fields than is even theoretically possible with Nb. Depositing such a coating on a complex geometry, such as a TESLA cavity, is likely to yield inhomogeneous coatings with potentially measurable impact on quantities of interest, such as the quality factor. We attempt to estimate these impacts through finite element (FE) simulations.

In our work, we model the SIS multilayer by reducing it to a surface impedance using a first-order Leontovich boundary condition [3]. This surface impedance appears in the boundary condition of the complex eigenvalue problem for the eigenmodes of the SRF cavity, which we solve via the FE method. We further treat the coating thickness as a Gaussian random field, obtained by solving a 2D FE problem on the boundary [4], which yields a spatially inhomogeneous surface impedance. By sampling Gaussian random fields we generate a normal distribution for the quality factor. We perform these simulations for a single-cell TESLA cavity and repeat them for different correlation lengths in the Matérn kernel of the Gaussian random field. We then compare the distributions of the quality factors by statistical methods.

References

[1] Valente-Feliciano, A.-M. (2016). Superconducting RF materials other than bulk niobium: a review. Superconductor Science & Technology, 29(11), 113002. DOI: 10.1088/0953-2048/29/11/113002

[2] Gurevich, A. (2006). Enhancement of rf breakdown field of superconductors by multilayer coating. Applied Physics Letters, 88(1), 012511. DOI: 10.1063/1.2162264

[3] Kubo, T. (2017). Multilayer coating for higher accelerating fields in superconducting radio-frequency cavities: a review of theoretical aspects. Superconductor Science & Technology, 30(2), 023001. DOI: 10.1088/1361-6668/30/2/023001

[4] Koh, K. J., & Cirak, F. (2023). Stochastic PDE representation of random fields for large-scale Gaussian process regression and statistical finite element analysis. Computer Methods in Applied Mechanics and Engineering, 417(116358), 116358. DOI: 10.1016/j.cma.2023.116358

A fully monolithic formulation for multiphase fluid-structure interaction

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Submission type: Presentation

Presenter: Marc Hirschvogel (Division 2.2 Process Simulation, Bundesanstalt für Materialforschung und -prüfung (BAM))

Computational modeling of fluid-structure interaction as well as multiphase fluid flow represent highly relevant approaches for insights into many engineering systems, but few works have thoroughly studied the combined effects from a numerical perspective. To-date approaches are either partitioned schemes [3] or fully Eulerian [2], limiting solver robustness or resolution of the fluid-solid interface. We present a first unified and monolithic finite element approach to multiphase fluid-structure interaction, where the flow is described by a coupled five-field Cahn-Hilliard Navier-Stokes equation system [1] in Arbitrary Lagrangian-Eulerian (ALE) description, and the structure is governed by finite strain elastodynamics. Unknowns of the resulting six-field system are fluid velocities, pressures, phase field, chemical potential, domain displacements, and structural deformation. Coupling of fluid and solid is achieved with a monolithic Neumann-Dirichlet scheme, where the structure is constrained by fluid kinematics and the fluid receives the reaction forces, circumventing the introduction of a Lagrange multiplier. Avenues for the effective solution of the resulting system are shown, and applications to elasto-capillarity and bubble-membrane interactions in electrolyzers are demonstrated.

References

[1] Brunk, A. and ten Eikelder, M. F. P. (2026). A simple, fully-discrete, unconditionally energy-stable method for the two-phase Navier-Stokes Cahn-Hilliard model with arbitrary density ratios. Journal of Computational Physics, 548:114558. DOI: 10.1016/j.jcp.2025.114558

[2] Valizadeh, N., Zhuang, X., and Rabczuk, T. (2025). A monolithic finite element method for phase-field modeling of fully Eulerian fluid-structure interaction. Computer Methods in Applied Mechanics and Engineerings, 435:117618. DOI: 10.1016/j.cma.2024.117618

[3] Zheng, X. and Karniadakis, G. E. (2016). A phase-field/ALE method for simulating fluid-structure interactions in two-phase flow. Computer Methods in Applied Mechanics and Engineering, 309:19-40. DOI: 10.1016/j.cma.2016.04.035

Nonlinear Inversion Framework - Magnetic Resonance Elastography (MRE)

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Submission type: Presentation

Presenter: Henrik Palme (KTH - Royal Institute of Technology, Department of Biomedical Engineering and Health Systems, Stockholm, Sweden)

Magnetic Resonance Elastography (MRE) is an emerging technique to map the material properties of soft tissue using Magnetic Resonance Imaging (MRI). While the method has been clinically implemented to assess the liver, further developments have to be made to expand the technique to other organs, such as the brain (Pepin et al., 2022). In brain MRE, the material field has been linked to neurodegenerative diseases, such as Parkinson's disease (Olsson et al., 2025) and Alzheimer's (Murphy et al., 2011). One clear obstacle toward the clinical implementation is the inversion process, where the displacement field of the tissue is used to generate the material parameters. While various methods are utilised to conduct the inversion, what is commonly seen as the most robust is nonlinear inversion (NLI) using the finite element method. NLI utilises a subzone-based iterative optimisation process to, by consecutive forward simulations, reach a material field that most closely creates a corresponding displacement field that matches the measured data (McGarry et al., 2012). This method is, however, very computationally expensive, requiring CPU clusters. Furthermore, the current implementation of NLI utilises Fortran and is not open source. For the further development of brain MRE, it is clear that improvements have to be made to the inversion methods. An open source framework utilising FEniCS is therefore being developed, both as a transparent alternative to the current implementation of NLI, but also as a way to generate physically accurate synthetic displacement fields from phantom geometries and material fields. This would enable both the validation of other inversion techniques and new ways of regularising deep learning methods using synthetically generated fields. In its current stage, the framework is limited to smaller phantom geometries and 2D inversion. Full 3D inversion of brain MRE data is currently being developed.

References

McGarry, M. D., Van Houten, E. E., Johnson, C. L., Georgiadis, J. G., Sutton, B. P., Weaver, J. B., & Paulsen, K. D. (2012). Multiresolution MR elastography using nonlinear inversion. Medical physics, 39(10), 6388-6396. DOI: 10.1118/1.4754649

Murphy, M. C., Huston, J., 3rd, Jack, C. R., Jr, Glaser, K. J., Manduca, A., Felmlee, J. P., & Ehman, R. L. (2011). Decreased brain stiffness in Alzheimer's disease determined by magnetic resonance elastography. Journal of magnetic resonance imaging : JMRI, 34(3), 494-498. DOI: 10.1002/jmri.22707

Olsson, C., Skorpil, M., Svenningsson, P., & Moreno, R. (2025). Effects of Parkinson's disease on mechanical and microstructural properties of the brain. NeuroImage. Clinical, 48, 103857. DOI: 10.1016/j.nicl.2025.103857

Pepin, K. M., Welle, C. L., Guglielmo, F. F., Dillman, J. R., & Venkatesh, S. K. (2022). Magnetic resonance elastography of the liver: everything you need to know to get started. Abdominal radiology (New York), 47(1), 94-114. DOI: 10.1007/s00261-021-03324-0

In-silico drug delivery in the human brain

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Submission type: Presentation

Presenter: Cécile Daversin Catty (Simula Research Laboratory)

Characterizing brain fluid dynamics and solute transport is a key ingredient in refining treatment delivery for neurological diseases. Despite extensive research, the underlying mechanisms are still subject to debate, and drug dosages are not yet tailored at the individual level. Biomedical computational models represent an innovative shift towards personalized adjustment of therapeutic dosage. These models must address the complexity and multi-scale aspect of the brain, and account for the large variability between individuals. Yet, this approach raises multiple challenges with respect to patient-specific parameterization, multiphysics coupling, and computational cost.

An inherent challenge lies in the processing and analysis of raw MRI data [1], as well as the subsequent generation of anatomically accurate, personalized meshes [2, 3]. We showcase a pipeline capable of generating FEniCSx-ready personalized meshes, delineating the critical stages of the workflow. Drug delivery predictions based on these meshes are derived from a 3D transport model, governed by an advection-diffusion equation and coupled with a Stokes model predicting the cerebrospinal fluid flow [4]. We present patient-specific simulations, with particular emphasis on comparison against MRI-derived quantities of interest.

References

[1] MRI-toolkit, https://github.com/scientificcomputing/mri-toolkit

[2] WildMeshing, https://github.com/wildmeshing

[3] PyVista, https://github.com/pyvista/pyvista

[4] Marius Causemann et al. (2025) In-silico molecular enrichment and clearance of the human intracranial space. bioRxiv preprint 2025.01.30.635680, version 1

Data-integrated simulations of solute transport in the rodent brain using FEniCSx

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Submission type: Presentation

Presenter: Andreas Solheim (Simula)

The glymphatic theory describes the extra-vascular transport and clearance of solutes in the brain. This solute transport is mediated by a set of barriers which are only partially understood. In this work, we emphasize the role of the pia membrane, the interface between the brain parenchyma and the cerebrospinal fluid (CSF), in determining solute transport in brain tissue. We use finite element simulations using FEniCSx with previously published data of tracer injection in 8 rats recorded by Single-Photon Emission Computed Tomography (SPECT) (Sigurdsson et al 2022). Parameter sweeps are performed using a quasi-Monte Carlo approach to estimate transport properties of this tracer, validating model outcomes with data recorded experimentally. Simulations use experimental data on the boundary of the domain, and we explore how the formulation of the boundary condition changes the transport properties predicted by the model. We show how FEniCSx can be combined with experimental data to provide insight into biological models and potentially inform new experiments.

References

Sigurdsson, et al., A spect-based method for dynamic imaging of the glymphatic system in rats. J. Cereb. Blood Flow & Metab. 43, 1153-1165 (2023) PMID: 36809165.

Poromechanics to Investigate the Impact of Mechanical Loading on Human Skin Micro-Circulation

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Submission type: Presentation

Presenter: Thomas Lavigne (Laboratoire de mécanique des solides, CNRS, École polytechnique, Institut Polytechnique de Paris, Palaiseau, France)

Human skin functions as a complex multi-scale and multi-phase system where moving fluids significantly influence mechanical and biological responses. Effective management of skin injuries, such as pressure ulcers (PU), requires a deep understanding of this structural composition and mechanical behavior, particularly given that between 9% and 20% of hospitalized patients in Europe are affected (Vanderwee et al., 2007). This research introduces a hierarchical two-compartment poromechanical model that accounts for fluid distribution within the interstitium and blood micro-circulation. The model is grounded in a hierarchical porous media framework (Lavigne et al., 2025) that conceptualizes the interstitium as a biphasic system, distinguishing between the characteristic timescales of cells and interstitial fluid.

Experimental evaluation was performed using Laser Doppler Flowmetry (LDF) on 11 healthy volunteers. Controlled loads were applied directly to the skin via a specialized pivotmeter device (Fromy et al., 1998) to investigate ischaemic and hyperaemic responses. All numerical simulations were conducted using the open-source software FEniCSx v0.9.0.

Results demonstrated that while absolute LDF values vary between individuals, relative responses to load-induced ischaemia and post-occlusive reactive hyperaemia (PORH) are comparable across sexes when normalized to basal blood flow. Sensitivity analysis identified Young's modulus and the vessel permeability exponent as dominant parameters governing the micro-circulatory response. This work successfully demonstrates the model's qualitative ability to replicate in vivo hemodynamic responses.

References

Vanderwee, K., Clark, M., Dealey, C., Gunningberg, L., & Defloor, T. (2007). Pressure ulcer prevalence in Europe: a pilot study. Journal of evaluation in clinical practice, 13(2), 227-235. DOI: 10.1111/j.1365-2753.2006.00684.x

Fromy, B., Abraham, P., & Saumet, J. L. (1998). Non-nociceptive capsaicin-sensitive nerve terminal stimulation allows for an original vasodilatory reflex in the human skin. Brain research, 811(1-2), 166-168. DOI: 10.1016/s0006-8993(98)00973-1

Lavigne, T., Urcun, S., Fromy, B., Josset-Lamaugarny, A., Lagache, A., Suarez-Afanador, C. A., Bordas, S. P. A., Rohan, P. Y., & Sciumè, G. (2025). Hierarchical Poromechanical Approach to Investigate the Impact of Mechanical Loading on Human Skin Micro-Circulation. International journal for numerical methods in biomedical engineering, 41(7), e70066. DOI: 10.1002/cnm.70066

Contact modeling using the Third Medium Contact (TMC) method: toward the application of a femoral stem insertion inside a bone cavity

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Submission type: Presentation

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Contact problems are among the most challenging in computational mechanics. Classical approaches such as the penalty method, the Augmented Lagrange Multiplier method [4], and the Nitsche method [3] require the definition of source-destination contact pairs and distance searches to evaluate the gap function. These approaches are not straightforward to implement in FEniCSx. From a different perspective, the so-called Third Medium Contact (TMC) method [6] introduces a fictitious medium between the contacting bodies with extremely low stiffness. From a strain energy barrier viewpoint, the energy remains negligible when the bodies are separated and increases abruptly as they approach contact. However, the method remained limited due to numerical issues caused by extremely distorted elements. Recently, some works have addressed these difficulties through strain-gradient-type regularization [2] or relaxed mixed formulations with low-order elements [5], making the method practically usable. From an implementation standpoint in FEniCSx, the TMC approach is relatively straightforward because it bypasses contact pair searches and instead relies directly on energy minimization, leveraging the FEniCSx's native automatic differentiation support. In this work, we focus on the simulation of total hip arthroplasty using a cementless approach, commonly employed in orthopedic surgery, where friction is the primary mechanism responsible for implant stability. To model this problem, we consider an elastodynamic formulation using the Newmark method for time discretization in a simplified axisymmetric geometry representing the insertion of a femoral stem into a bone cavity. Using the analogy between plasticity and friction, the formulation of a friction law is introduced in the TMC method which is implemented with the help of the dolfinx_materials library [1]. Finally, a parametric study will be performed, and the results will be contrasted with the Nitsche method provided by COMSOL.

References

[1] Bleyer, J. (2025). dolfinx_materials: A Python package for advanced material modelling, DOI: 10.5281/zenodo.13882183

[2] Bluhm, G. L., Sigmund, O. & Poulios, K. (2021). Internal contact modeling for finite strain topology optimization. Computational Mechanics, 67, 1099-1114. DOI: 10.1007/s00466-021-01974-x

[3] Chouly, F., Hild, P. & Renard, Y. (2023). Finite Element Approximation of Contact and Friction in Elasticity (Birkhäuser Cham, AMMA, 48, ACM). DOI: 10.1007/978-3-031-31423-0

[4] Laursen, T. A. (2003). Computational contact and impact mechanics: fundamentals of modeling interfacial phenomena in nonlinear finite element analysis (Springer Science & Business Media, 1). DOI: 10.1007/978-3-662-04864-1

[5] Wriggers, P., Korelc, J. & Junker, Ph., (2025). A Third Medium Approach for Contact Using first and Second Order finite Elements. Computer Methods in Applied Mechanics and Engineering, 436, 117740. DOI: 10.1016/j.cma.2025.117740

[6] Wriggers, P., Schröder, J., Schwarz, A., (2013). A finite element method for contact using a third medium. Computational Mechanics, 52(4), 837-847. DOI: 10.1007/s00466-013-0848-5

Inferring Human Intracranial CSF Absorption Sites via Inverse Modelling of Protein Transport

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Submission type: Presentation

Presenter: Marius Causemann (Simula Research Laboratory)

The clearance of metabolic waste products and the distribution of therapeutic agents within the central nervous system are critically dependent on the circulation of cerebrospinal fluid (CSF). However, the precise anatomical locations of CSF production and absorption in the human brain remain subjects of ongoing investigation.

Recent clinical research [Storås et al. 2025] and open tracer evolution datasets [Riseth et al. 2026] utilized high-resolution T1 mapping (R1) to demonstrate significant regional solute gradients in the CSF of healthy brains, where R1 in the subarachnoid space (SAS) correlates strongly with intrinsic protein concentration. These findings suggest that quantitative R1 maps can serve as a reliable, non-invasive proxy for protein distribution within the human intracranial space.

Building on this physiological insight, we present a computational framework to infer CSF absorption pathways by integrating mathematical modeling with clinically acquired R1 imaging data. We model the transport of proteins within the CSF-filled spaces using a coupled fluid flow and mass transport model. First, we generate a set of basis velocity fields by solving the steady-state Stokes equations, where each field represents a distinct hypothesis for a specific absorption site. Second, we model the transport of proteins using a stationary advection-diffusion equation driven by a linear combination of these basis fields.

To identify the active absorption sites, we formulate the inverse problem as a PDE-constrained optimization. We define a objective function that minimizes the mismatch between the computed equilibrium protein concentration and the measured R1 data field. By solving for the optimal combination of basis flows that best reproduces the observed protein gradients, this approach provides a mathematical tool for non-invasively characterizing intracranial fluid dynamics and validating physiological transport hypotheses.

References

Storås, Tryggve Holck, et al. "T2-Weighted T1 Mapping and Automated Segmentation of CSF: Assessment of Solute Gradients in the Healthy Brain." Journal of Magnetic Resonance Imaging (2025).

Riseth, Jørgen N., et al. "Human brain MRI data of intrathecally injected tracer evolution over 72 hours for data-integrated simulations." Scientific Data (2026).

Material model discovery with FEMU in FEniCSx

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Submission type: Presentation

Presenter: Saeid Ghouli (ETH Zurich)

Our group has developed a computational framework based on sparse regression, denoted as EUCLID (Efficient Unsupervised Constitutive Law Identification & Discovery; Flaschel et al., 2021, 2023), to perform automated discovery of material laws out of a library of candidate functions. The loss function in the inverse problem has been formulated based on unbalanced internal and boundary forces computed from full-field displacement and global force data, similar to identification strategies such as the equilibrium gap and the virtual fields methods. The computation of the loss function thus involves the numerical differentiation of the experimental displacement data to obtain the strains, an operation that is notoriously sensitive to noise. In the present work, we aim at combining the discovery process of EUCLID based on sparse regression with a loss function based on the displacement field and reaction forces, in the spirit of the so-called finite element model updating (FEMU) technique. Exploiting the experimentally measured displacement data directly (without differentiation) provides a more noise-robust discovery algorithm; however, the new inverse problem becomes non-convex, requiring robust optimisation techniques, and involves finite element analyses at each iteration, requiring an efficient implementation. Also, since there are many models in the candidate library, we need to rely on automatic differentiation to avoid deriving all linearisations by hand. For all these reasons, we implement FEMU (forward and backward problems) via FEniCSx. In this talk, we present the key points of the implementation, illustrate the solution to the non-convex, non-linear inverse problem, and highlight the potential of our new approach.

References

Flaschel, M., Kumar, S., & De Lorenzis, L. (2021). Unsupervised discovery of interpretable hyperelastic constitutive laws. Computer Methods in Applied Mechanics and Engineering, 381, 113852. DOI: 10.1016/j.cma.2021.113852.

Flaschel, M., Kumar, S., & De Lorenzis, L. (2023). Automated discovery of generalized standard material models with EUCLID. Computer Methods in Applied Mechanics and Engineering, 405, 115867. DOI: 10.1016/j.cma.2022.115867.

Finite-Pressure Anisotropic Hyperelasticity in FEniCSx: A Phenomenological Approach for Shock Physics

Paul Bouteiller

CEA DAM

Submission type: Presentation

Presenter: Paul Bouteiller (CEA)

We present a finite-strain anisotropic hyperelastic model suited to materials subjected to high-amplitude shock loading, where hydrostatic pressures far exceed deviatoric stresses. The formulation relies on a multiplicative decomposition of the deformation gradient, separating the pressure-induced lattice distortion from the residual isochoric distortion. The Helmholtz free energy is then expanded as a quadratic form in the intermediate Green-Lagrange strain, cleanly decoupling three independently calibrated contributions: the equation of state, the geometric lattice distortion, and the pressure-dependent isochoric stiffness tensor. Each component can be identified directly from DFT calculations, molecular dynamics simulations, or Diamond Anvil Cell experiments. The model naturally covers all crystallographic symmetry classes from cubic to triclinic through a gauge freedom in the isochoric stiffness.

The constitutive law is implemented in the open-source code CHARON (https://github.com/PaulBouteiller/Charon), built on FEniCSx. The total Lagrangian formulation and the UFL automatic differentiation framework allow the free energy to be transcribed almost directly from its mathematical expression, yielding both the stress and the consistent tangent without any additional derivation effort from the user.

Validation is carried out on cubic copper, tetragonal PETN, orthorhombic alpha-RDX. Polycrystal homogenisation on 500-grain meshes generated with NEPER accurately reproduces the pressure dependence of the macroscopic shear modulus.

High-performance solid mechanics simulations in FEniCSx

Musa Choudhury, Garth Wells

University of Cambridge

Submission type: Presentation

Presenter: Musa Choudhury (University of Cambridge)

There is increasing demand for high-performance, nonlinear solid mechanics solvers, with simulations of component systems now being routinely three-dimensional and larger in scale. Previously we demonstrated a high-performance solid mechanics library in DOLFINx interfacing Abaqus style UMAT subroutines which are standardly used for writing constitutive algorithms. This capability allows us to run complex constitutive algorithms such as crystal plasticity models on massively scalable HPC hardware. A significant bottleneck becomes the memory required to deploy these models. We also demonstrated the performance gains when using an O(n) iterative solver as opposed to O(n2) direct solver as is commonly used in industry. The problem now becomes ensuring that the iterative solver converges for the class of problems we are interested in. To overcome this, efficient and robust preconditioning techniques need to be developed. We analyse the algebraic multigrid preconditioner with the conjugate gradient method for solid mechanics problems including plasticity. We will then present ways to improve its performance to achieve a robust solver with respect to mesh refinement and material parameters.

Flexibility Method for Contact Mechanics in FEniCSx. Finite Element Construction and Hierarchical Compression of Compliance Operators.

Yahya Boye[1], Jérémy Bleyer[2], Stéphanie Chaillat[3], V. A. Yastrebov[1]

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Submission type: Presentation

Presenter: Yahya Boye (Mines Paris PSL)

Contact problems in solid mechanics are classically formulated by the Moreau-Signorini conditions, expressing non-penetration, non-adhesion, and complementarity. This leads to a variational inequality with a unique solution (Duvaut & Lions, 1976). In practice, these problems are often solved with FEM using penalty, Lagrange multiplier, or augmented Lagrangian methods (Laursen, 2002). Although effective, such approaches introduce user -defined parameters or additional unknowns and are intrusive, since they require modifications of the tangent matrix structure.

In this work, we investigate the flexibility method originally introduced by Francavilla and Zienkiewicz (1975). This approach reformulates the contact problem using the Neumann-to-Dirichlet operator restricted to the contact interface. This operator maps contact tractions to surface displacements, allowing the problem to be recast as a linear complementarity problem (Cottle et al., 1992). The contact forces are solved as an auxiliary problem and then applied as Neumann boundary conditions in the FEM model.

A main limitation of this approach is the dense compliance operator, which leads to poor scalability. To address this, we use hierarchical matrices (Hackbusch, 2015), which compress low-rank far-field interactions and reduce storage and matrix-vector costs.

We also investigate several strategies to construct the compliance operator, including sampling, Schur complement, and BEM. A key advantage of the method is its non-intrusive character: contact is solved as an auxiliary problem without modifying the finite element formulation. The method is implemented in an open-source framework based on FEniCSx. Future work will address frictional contact and nonlinear materials

References

[1] Duvaut, G., & Lions, J.-L. (1976). *Les inéquations en mécanique et en physique*. Springer. DOI: 10.1007/978-3-642-66165-5

[2] Laursen, T. A. (2002). *Computational contact and impact mechanics*. Springer. DOI: 10.1007/978-3-662-04864-1

[3] Francavilla, A., & Zienkiewicz, O. C. (1975). A note on numerical computation of elastic contact problems. *International Journal for Numerical Methods in Engineering, 9*(4), 913-924. DOI: 10.1002/nme.1620090410

[4] Cottle, R. W., Pang, J.-S., & Stone, R. E. (1992). *The linear complementarity problem*. SIAM. DOI: 10.1137/1.9780898719000.fm

[5] Hackbusch, W. (2015). *Hierarchical matrices: Algorithms and analysis*. Springer. DOI: 10.1007/978-3-662-47324-5

Computational Shakedown Analysis with FEniCSx: Conic Optimization for Large-Scale FEM

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Computational Applied Mechanics (University of Wuppertal)

Submission type: Presentation

Presenter: Lavakumar Veludandi (Computational Applied Mechanics, University of Wuppertal)

Shakedown analysis offers a robust framework for assessing the long-term structural integrity of materials under variable loading. Classical lower-bound approaches, based on Melan's Theorem, formulate this as a constrained optimization problem to determine the maximum safe load multiplier while satisfying equilibrium and yield criteria. However, discretizing these continuous variational inequalities into tractable large-scale finite element models remains computationally challenging due to the enormous number of resulting discrete constraints.

We present a modular, extensible computational framework for lower-bound shakedown analysis integrated natively within the FEniCSx ecosystem. This framework couples FE formulations with large-scale conic optimization. By leveraging the low-level assembly capabilities of FEniCSx, the approach maintains explicit control over constraint sparsity, enabling the construction of large block-sparse matrices suitable for high-resolution discretizations. The framework provides a flexible solver interface allowing users to select from high-performance backends like MOSEK or Gurobi, or integrate custom solvers, which facilitates direct result comparison and adaptable solution strategies while preserving the automation and versatility of the FEniCSx workflow.

Validation against benchmark problems confirms the accuracy and robustness of the proposed framework. The methodology is further applied to structures exhibiting heterogeneous and anisotropic material behavior under multidimensional loading. The approach efficiently computes structural shakedown limits and demonstrates the potential of combining automated finite element frameworks with advanced conic optimization for scalable structural limit analysis across diverse material systems.

References

François, A., Abdelkader, H., An, L. T. H., Said, M., & Tao, P. D. (2007). Application of lower bound direct method to engineering structures. Journal of Global Optimization, 37(4), 609-630. DOI: 10.1007/s10898-006-9069-1;

Simon, J.-W., & Weichert, D. (2011). Numerical lower bound shakedown analysis of engineering structures. Computer Methods in Applied Mechanics and Engineering, 200(41-44), 2828-2839. DOI: 10.1016/j.cma.2011.05.006;

Simon, J.-W., & Weichert, D. (2011). Shakedown analysis with multidimensional loading spaces. Computational Mechanics, 49(4), 477-485. DOI: 10.1007/s00466-011-0656-8;

Convergence Study on Polycrystalline Materials

Donald Boyce, Paul Shade

Cornell University, Air Force Research Laboratory, Cornell University, Air Force Research Laboratory

Submission type: Presentation

Presenter: Donald Boyce (Cornell University)

Polycrystalline materials possess distinct material properties at the individual grain scale, creating jumps (discontinuities) in properties at grain boundaries. This leads to limitations in solution regularity and global convergence rates of finite element calculations. This study examines convergence rates on a grain-by-grain basis, focusing on the convergence of grain-averaged quantities. We utilize conductive heat transfer and linear elasticity as model problems to investigate the relationship between relative errors and grain size.

Variational Solvers for Irreversible Evolutionary Systems: Orchestration

Andrés A. León Baldelli ; Pierluigi Cesana

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Submission type: Presentation

Presenter: Andrés A León Baldelli (∂'Alembert / CNRS, Sorbonne Université)

Irreversible variational models for fracture and damage are challenging for numerical simulation: the problem is non-convex, constrained, and may admit multiple admissible paths. The predictive power of simulations depends not only on the model itself, but on how equilibria are identified, tracked, and selected.

In this talk, I present a modular computational framework for the simulation of irreversible evolutionary systems, built on FEniCSx and integrating three nonlinear variational solvers addressing complementary aspects of the problem: constrained equilibrium, loss of uniqueness of evolution paths, and a rigorous criterion to assess the stability of computed states.

Rather than treating these components independently, I focus on their orchestration. At each load increment, the system is first brought to constrained equilibrium, then interrogated for bifurcation, and finally tested for stability. This structured interaction between first- and second-order information provides a strategy way to navigate the multiplicity of solutions and to track physically meaningful evolution paths.

The framework is implemented in Python using DOLFINx, PETSc, and SLEPc, leveraging automatic differentiation and scalable linear algebra.

More broadly, with this approach I illustrate how variational solvers can be composed into a transparent and extensible workflow for irreversible systems driven by energetic principles, providing a scalable foundation for more complex models, where additional internal variables and solver components can be integrated in a systematic manner, facilitating the implementation of experimental validation pipelines.

References

A. A. León Baldelli and P. Cesana. Variational solvers for irreversible evolutionary systems. Journal of Open Source Software, 9(104), 2024.

Implementation of the TiNiest Tensor de Rham subcomplex on 3D Hybrid Meshes

Johnny Vogels

Affiliation unavailable

Submission type: Presentation

Presenter: Johnny Vogels

The differential resolution is given by the de Rham complex. In three dimensions, the operators gradient, curl, and divergence constitute the differentials that map scalars, covectors, vectors, and densities to the next level in the sequence of the complex.

Once the domain is equipped with a mesh, discrete subcomplexes with commuting projections can be constructed, forming the foundation for stability using Finite Element Exterior Calculus (FEEC) with Hodge-Laplacian or Hodge-Dirac formulations. This becomes essential when considering mixed formulations where (co-)vectorial quantities get endowed with degrees of freedom.

On simplicial meshes, either full or trimmed spaces of specific polynomial orders are used. This choice determines the relative accuracy of quantities in mixed formulations. For hybrid meshes - comprising tetrahedra, hexahedra, prisms, and pyramids - additional choices arise. With full spaces, one uses the serendipity subcomplex. We are exploring further options. For trimmed (Ciarlet-)elements, three subcomplexes with successively increasing accuracy under non-affine mappings are available: trimmed serendipity, TiNiest Tensor (TNT), and full tensor-product (Q-).

We apply the mathematically required corrections to the formulations of Cockburn and Fu (2017). On this basis, we present the first implementation of the complete TiNiest Tensor De Rham subcomplex on 3D hybrid meshes. This is also the first implementation of any 3D hybrid-mesh subcomplex in symfem and basix.

This enables simulations of basic 3D processes on hybrid meshes with (co-)vectorial degrees of freedom such as electromagnetism.

Keywords: TiNiest Tensor, transition elements, prism, pyramid, de Rham, conformal, Sobolev complex, FEEC, commuting projections, Hodge-Laplacian, Hodge-Dirac, symfem, basix, Ciarlet, saddle-point systems, mixed methods, vectorial quantities, electromagnetism, element families, discrete subcomplexes, hybrid meshes, serendipity spaces, trimmed spaces

References

Cockburn, B., Fu, G. (2017). A systematic construction of finite element commuting exact sequences. SIAM Journal on Numerical Analysis, 55(4), 1650-1688.

AN IMPLEMENTATION OF THE DISTRIBUTED TWO SCALES METHOD FOR LINEAR ELASTICITY PROBLEMS BASED ON FENICSX AND PETSC

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Submission type: Presentation

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The distributed two-scale method, proposed in Salzman and Moës (2023), has been reimplemented with FEniCSx and PETSc libraries as an independent Python/C++ library. This method uses enrichment techniques to compute a fine-scale discretized problem without solving the associated linear algebra system. Instead, a set of small problems covering the main zone of interest are solved, and their solutions are used as enrichment functions for a coarser scale problem. Solving this coarse-scale problem is expected to be relatively cheap. The result of this coarse system can then be projected at the fine scale. Recomputing the small problems with these new boundary conditions leads to loops between the two scales and converges to the solution of the fine scale problem.

This talk gives the state of the art of this ongoing work. It discusses the various approaches used to implement this method, and, in particular, the choices made on which parts are treated by FEniCSx, by PETSc, and coded in the library. Comparison with seminal implementation on some computation shows that the current version of the library is providing same numerical results. Some other test cases also confirm the correct implementation in the context of parallel computation with use of multi point constraints, thanks to dolfinx mpc.

Finally, an extra numerical analysis of the use of the distributed two-scale method as a preconditioner illustrates the interest of coding in the FEniCSx ecosystem, which offers a high level of flexibility for prototyping in research.

References

Salzman, A. & Moës, N. (2023) A two-scale solver for linear elasticity problems in the context of parallel message passing. Computer Methods in Applied Mechanics and Engineering. DOI: 10.1016/j.cma.2023.115914

Adjoint-accelerated inverse modelling of acoustic cross-talk in piezoelectric inkjet printheads

Javier Lorente-Macías

University of Cambridge

Submission type: Presentation

Presenter: Javier Lorente-Macías (University of Cambridge)

We develop an adjoint-accelerated inverse modelling framework to generate accurate, interpretable models of the internal fluid dynamics in drop-on-demand piezoelectric inkjet printheads. From Laser Doppler Vibrometry (LDV) measurements of meniscus oscillations, we jointly infer internal acoustic flow parameters and reconstruct the acoustic flow field within the printhead. Despite the sparsity of the data, the inferred model reproduces the observed dynamics with high accuracy. The reconstruction further reveals cross-talk mechanisms not directly accessible experimentally, including direct acoustic excitation between adjacent channels, acoustic wave propagation through the printhead structure, and coupling via the shared ink manifold. Building on these insights, we derive reduced-order models (ROMs) describing the internal printhead dynamics. We then use Bayesian inference for parameter estimation, uncertainty quantification, and model selection. The resulting ROM is both interpretable and predictive, capturing the essential dynamics with a small number of physical parameters. Finally, we demonstrate accurate extrapolation to unseen printheads, highlighting the robustness of the approach. This work establishes a PDE-constrained inverse modelling approach for complex acoustic channel dynamics in industrial applications, enabling predictive simulation and reducing reliance on trial-and-error experimental testing.

References

[1] Adjoint-based optimization of open-loop control for microfluidics of an inkjet printhead (Javier Lorente-Macías; Matthew P. Juniper), The Journal of the Acoustical Society of America.

[2] Fundamental Fluid Dynamics Challenges in Inkjet Printing (Detlef Lohse), Annual Review of Fluid Mechanics.

FEniCS Post-processing with ParaView

Louis Gombert

Kitware

Submission type: Presentation

Presenter: Louis Gombert (Kitware)

In this presentation, we introduce the open source post-processing toolkit ParaView [0]. ParaView is an industry-standard graphical application that also provides a Python API, used for interactive 3D data exploration and analysis. ParaView can efficiently process data in parallel up to billions of cells, either from files on disk, or from in-memory data using the Catalyst framework. We show the different ways ParaView can be deployed: as a standalone graphical application, using remote rendering on a supercomputer, on the web through the Python interface. Then, we show how ParaView can be connected to FEniCS simulations, how its data model binds to ParaView data structures, and how it can produce publication-ready image and video renderings easily. We also discuss prospective topics regarding higher-order element support and advanced filters usage.

References

[0] Ahrens, J., Geveci, B., & Law, C. (2005). Paraview: An end-user tool for large data visualization. The visualization handbook, 717(8).

The Digital Math framework as a robust and general scientific basis for society - advancements in the Adaptive Euler breakthrough for aerodynamics and DigiMat educational program

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Submission type: Presentation

Presenter: Johan Jansson (KTH Royal Institute of Technology)

Today we see large-scale threats to the scientific basis of modern society, which often contain elements of attacks on the concept of truth. We develop the Digital Math framework - where FEniCS is a key component - to form a robust general scientific basis for research and education, and a counter to the threats to science by guaranteeing reproducibility and falsifiability.

All of natural science and engineering is formulated as mathematical equations - models, and the Digital Math framework provides powerful and at the same time simple and reproducible methods and tools for predicting/solving the models, and pedagogical frameworks for learning the methods and models. Key components are the FEniCS framework for automated computational modeling and the DigiMat motivational educational framework.

We here contribute advancements in the Adaptive Euler methodology for first-principles reliable and efficient aerodynamics - solving the Grand Challenge of aerodynamics, and the Unified Continuum methodology for general and robust continuum modeling such as fluid-structure interaction (FSI).

We present specific developments of Adaptive Euler and DigiMat:

1. A simple and general Adaptive Euler formulation validated in the High Lift, Drag and Aeroelasticity Prediction workshops, including compressible phenomena such as shocks, resolving the Grand Challenge of aerodynamics with fast and cheap first-principles aerodynamics simulation.

2. Development toward a general and first-principles Adaptive Euler Unified Continuum FSI formulation with the ability to predict aeroelastic phenomena such as: flutter and buffeting (wake-tail and transonic).

3. Industrial workflows and open model results developed together with SAAB and FOI.

4. DigiMat pedagogical learning activities developed in collaboration with the Technical Museum and IVA Royal Academy of Engineering Sciences Technology Leap.

For references and more details, see: http://digitalmath.tech/fenics2026-JohanJansson/

References

For references and more details, see:

http://digitalmath.tech/fenics2026-JohanJansson/

Generalized Stokes problem with pressure Dirichlet boundary conditions for heat transport

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Universitat Autònoma de Barcelona

Submission type: Presentation

Presenter: Xavier Cartoixà (Universitat Autònoma de Barcelona)

The Boltzmann Transport Equation (BTE) describes semiclassical transport by charge and heat carriers: electrons and phonons, respectively. After some manipulations and approximations, the BTE naturally yields Ohm's law and diffusion effects for electrons, and Fourier's law in the case of phonons. If some of the approximations are relaxed, it was shown by Guyer and Krumhansl that there appear some extra terms in Fourier's law which endow the heat flux with viscous properties. These "phonon hydrodynamics" effects manifest typically in the nanoscale, and with geometries where the flux lines are expected to have a curvature radius of the order of a hydrodynamic length. Mathematically, the steady-state Guyer-Krumhansl (GK) equations are analogous to the Generalized Stokes Problem (GSP) for an uncompressible fluid, with the temperature playing the role of the pressure, and the heat flux playing the role of the velocity field. The notable difference is that, while in the GSP the boundary conditions (BCs) are rarely expressed in terms of the pressure, in the GK equations it is natural to fix the temperature at, typically, hot and cold contacts, with the rest of the boundary under no- or partial-slip BCs. We will report our experience solving the GK equations with FEniCSx, successfully implementing weak forms of the pressure Dirichlet BC with the Nitsche method in the case of Poiseuille flow, where the velocity/heat flux is guaranteed to exit and reach the contacts orthogonally to the contact surface. However, more complicated geometries with the heat flux possibly exiting/arriving at the contact at an angle do not seem to lend themselves naturally to symmetric forms, leaving us with a nonsymmetric formulation. We will discuss uniqueness and stability considerations in this setting. We acknowledge financial support by Spain's Ministerio de Ciencia, Innovación e Universidades under project No. PID2024-161603NB-I00 (MICIU / AEI / 10.13039/501100011033 / FEDER, UE).

Electromagnetic Waveguide Input/Output Boundary Condition using SciFEM and FEniCSx

G. William Slade

Independent

Submission type: Presentation

Presenter: G. William Slade (Independent)

The ability to simulate the introduction and absorption of electromagnetic power at waveguide port boundaries is a fundamental challenge of computational electromagnetics. From the results of the internal Maxwell problem, this boundary model is used to compute a set of scattering parameters that define the performance of a microwave device. We approach the problem by augmenting the internal electromagnetic field problem (solved using the FEniCSx suite) with a boundary operator that models the exterior waveguide problem using a mode-matching approach. In order to construct this boundary operator, we use the SciFEM [1] real space facility to construct the required modal projection operators on the waveguide port boundaries. We cover the derivation and implementation of the method and describe how, in most cases, the sparseness of the finite element matrix operator is not noticeably degraded. Moreover, the boundary operator implementation directly yields the scattering parameters between ports and modes. Some illustrative examples are presented to demonstrate the efficacy of the method, as well as a comparison with a simpler, but less general impedance boundary condition method that we have long used [2]. We also briefly discuss how one might go about automatically generating the mode functions required by this method as well as linking this approach to the modeling of general open-region problems.

References

[1] Henrik Finsberg, & Jørgen Schartum Dokken. (2024). scientificcomputing/scifem: v0.2.2 (Version v0.2.2) [Computer software]. scientificcomputing/scifem: v0.2.2

[2] G. W. Slade, "Solving the Time-Harmonic Maxwell Equations in 3-D Using the FEniCSx Finite Element Package", Available at ResearchGate: https://www.researchgate.net/publication/367450648_Solving_the_Time-Harmonic_Maxwell_Equations_in_3-D_Using_the_FEniCSx_Finite_Element_Package_-_2023_Update

Towards fully programmable inflatable panels

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Submission type: Presentation

Presenter: Ofir Mirkin (CNRS - LIPhy)

Self-deployable materials are expected to have wide application potential, e.g., in robotics, medicine, and architecture. So far, all responsive materials have limited deformation possibilities, and researchers lack control over the six local degrees of freedom (three for the metric and three for the curvature). Here, we introduce a strategy to address this limitation.

We investigate how variations in mesoscale properties and applied pressure influence the deformation of inflatable elastic architected materials. Our structures feature a triangular network of interconnected cylindrical cavities, and we study how their density, eccentricity, and orientation control the in-plane deformation. To program out-of-plane curvature, we superimpose two layers with distinct cavity geometries, achieving a 2D generalization of the bilayer effect.

The design and analysis of these structures are driven by a computational framework implemented in FEniCSx. We leverage the library to perform periodic homogenization and simulate the non-linear hyperelastic behavior of the underlying incompressible elastomers. This high-fidelity numerical pipeline allows us to efficiently explore a vast design space of cavity geometries, that we then compare with experimental data.

This work paves the way for the inverse design of inflatable structures with fully programmable intermediate configurations and mechanical properties.

A Finite-Element Method for Fluctuating Navier–Stokes Equations

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Submission type: Presentation

Presenter: Dimitrios Gourzoulidis (Imperial College London)

Thermal fluctuations play a fundamental role in fluid dynamics, from Brownian motion at microscopic scales to activated processes such as cavitation, boiling, condensation, crystallisation, and hydrodynamic instabilities. While fluctuating Navier–Stokes equations have been studied extensively, finite-element discretisations for these systems remain comparatively underexplored. In this work, we present a finite-element framework for simulating compressible fluids governed by the fluctuating Navier–Stokes equations. The method preserves the fluctuation–dissipation balance at the discrete level by defining the stochastic forcing through its variational action, so that its covariance is proportional to the discrete viscous dissipation operator. Time integration is performed using a Crank–Nicolson scheme to ensure stability and accuracy. Numerical validation in one, two, and three spatial dimensions shows that the method correctly reproduces equilibrium fluctuation statistics for different mesh sizes and time steps.

References

Gourzoulidis, D., Gallo, M., Elkantassi, S., Kay, T., and Kalliadasis, S. A finite-element method for fluctuating Navier–Stokes equations. To be submitted.

Flow distortion generation using topology optimization

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Submission type: Presentation

Presenter: Langlet Nathan (DAAA, ONERA, Institut Polytechnique de Paris, 92190, Meudon, France)

This study investigates the use of topology optimization to automatically identify aerodynamic designs that accurately generate a specific target velocity profile. The methodology couples the steady-state RANS equations with the Spalart-Allmaras turbulence model and a relaxed eikonal equation for dynamic wall-distance computation, implemented using the FEniCSx library. To ensure clearly defined solid-fluid interfaces, the Topology Optimization of Binary Structures (TOBS) method is employed, utilizing integer linear programming via the CPLEX software. The approach is validated on the wake reproduction of a square cylinder at Re = 105, and further applied to flow separation control in a 90° sharp-angled bend at Re = 104. Results demonstrate that the framework successfully creates streamlined guide vanes that eliminate flow separation, maintaining target velocity profiles within a ±5% tolerance. The influence of initial guesses and optimization parameters is analyzed, highlighting the existence of multiple local minima and the validation of the method for complex turbulent flows. The governing equations include incompressible RANS with Brinkman term for solid modeling, and the S-A model with penalty for turbulence in solids. The objective is a weighted sum of velocity error at AIP and regularization. Numerical implementation uses Newton solver for nonlinear system and branch-and-bound for ILP. In the square case, objective reduces 380-fold, with geometry matching the target wake. For the bend, four cases show curved vanes, with convergence varying by initial guess, revealing TOBS efficiency in adding material. Conclusion suggests extensions to 3D compressible flows and TOBS-GT for better interface.

Finite-Element Modeling of Complex Inelasticity Using FEniCSx: Paper and Thermoplastics

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Submission type: Presentation

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Predicting the mechanical response of materials such as paper and thermoplastics is crucial for a wide range of applications, from sustainable to high-performance. Despite their different origins, both exhibit nonlinear behaviors under large deformations that challenge conventional computational frameworks. This work presents constitutive models for simulating their inelastic and coupled thermomechanical responses, implemented in FEniCSx.

For paper, an advanced large-deformation elasto-plastic model was developed to capture anisotropic behavior. The formulation distinguishes between in-plane and out-of-plane responses in the free energy and yield surface, reflecting the layered architecture of paper. Refinements address inconsistencies in initial yield and plastic strain ratios, while stiffness evolution due to densification during compression is included. Anisotropic hardening is modeled via coupled internal variables governing yield surface evolution.

For semi-crystalline thermoplastics, a visco-hyperelastic-plastic framework is introduced to describe the interaction between deformation, microstructure, and temperature. Based on a two-potential formulation, it ensures thermodynamic consistency while enabling viscoelastic relaxation, plastic flow, and thermal coupling. Crystallinity-dependent stiffening and self-heating effects capture temperature-driven transitions during loading. Both models are formulated using a specific intermediate configuration, offering physical clarity and algorithmic advantages.

The implementations rely on FEMExternalOperators and JAX for local solution schemes and automatic differentiation of consistent tangents; for thermoplastics, CUDA-based vectorization yields substantial speedups. Numerical studies confirm robustness and predictive capability. Overall, these developments highlight how FEniCSx supports rapid, reproducible modeling of complex inelastic materials within a unified open-source finite element environment.

References

- Cui, J., Liu, T., Valsecchi, M., Giersberg, M., Çelik, H., Simon, J.-W., Kumar, S., Petersen, J., & Fish, J. (2025). Thermodynamically consistent coupled chemo-thermo-mechanical model of interfaces in overmolded thermoplastic parts. Computer Methods in Applied Mechanics and Engineering, 447, 118359. DOI: 10.1016/j.cma.2024.118359

- Felder, S., Holthusen, H., Hesseler, S., Pohlkemper, F., Gries, T., Simon, J.-W., & Reese, S. (2020). Incorporating crystallinity distributions into a thermo-mechanically coupled constitutive model for semi-crystalline polymers. International Journal of Plasticity, 135, 102751. DOI: 10.1016/j.ijplas.2020.102751
- Holthusen, H., Rothkranz, C., Lamm, L., Brepols, T., & Reese, S. (2023). Inelastic material formulations based on a co-rotated intermediate configuration: Application to bioengineered tissues. Journal of the Mechanics and Physics of Solids, 172, 105174. Elsevier BV. DOI: 10.1016/j.jmps.2023.105174
- Kopic-Osmanovic, N., Prume, E., Felder, S., Kloppenburg, G., & Simon, J.-W. (2023). Modeling the anisotropic elasto-plastic material behavior of paper and paperboard at finite deformations. Conference Proceedings, 22, e202200293.
- Latyshev, A., Bleyer, J., Maurini, C., & Hale, J. (2025, October 8). Expressing general constitutive models in FEniCSx using external operators and algorithmic automatic differentiation. Journal of Theoretical, Computational and Applied Mechanics. Centre pour la Communication Scientifique Directe (CCSD). DOI: 10.46298/jtcam.14449
- Liu, T., Cui, J., Kantert, A., Tabib, M., Kopic, N., Valsecchi, M., Kumar, S. K., Fish, J., & Simon, J.-W. (submitted). Thermomechanically coupled visco-hyperelastic-plastic model with self-heating for thermoplastics.

Large scale random vibration analysis using FEniCS

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Submission type: Presentation

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Large-scale engineering structures, such as bridges, high-rise buildings, and offshore platforms, are often subjected to dynamic loads induced by environmental factors like earthquakes, wind, and traffic [1, 2]. Base excitation random vibration analysis plays a crucial role in assessing the structural response under stochastic excitations. This study investigates the dynamic behavior of large-scale structures under random base excitations using FEniCS. To simulate the real world, additional mass is applied to incorporate additional geometry of the assembly. Modal analysis is performed to determine the dynamic characteristics of the structure, including natural frequencies and mode shapes. The frequency response function is derived based on the system's natural frequency, damping ratio, and excitation frequency. Using the modal properties, the transfer function relating displacement response to base excitation is formulated. This expression incorporates the influence of mass, mode shapes, and transformation matrices. The acceleration response transfer function is obtained by considering the displacement response along with an additional transformation term. The power spectral density of the output response is then computed by multiplying the transfer function matrix with its conjugate transpose and the input power spectral density of the excitation. This formulation enables the evaluation of structural response characteristics under stochastic base excitations. Results of modal and random vibration analysis with and without additional mass are validated with an industry-standard commercial package.

References

[1] Saurabh, S., Gupta, A., Chowdhury, R., & Podugu, P. (2024). Robust topology optimization for transient dynamic response minimization. Computer Methods in Applied Mechanics and Engineering, 426, 117009.

[1] Saurabh, S., Gupta, A., Chowdhury, R., & Duddu, R. (2025). Robust topology optimization for transient dynamic response minimization. Reliability Engineering & System Safety, 264, 111440.

femOHL: A FEniCSx-based Framework for Structural Integrity and Resilience Assessment of Overhead Power Transmission Infrastructure

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Submission type: Presentation

Presenter: Daria Mesbah (Mews Labs)

In the context of climate change and increasingly severe extreme wind events, assessing the structural integrity of overhead transmission infrastructure under environmental loading has become a major strategic challenge for Transmission System Operators (TSOs), particularly for RTE in France, which operates the largest transmission network in Europe. The goal of this project is to develop a robust and accessible tool that can be used by decision-makers and field experts, without requiring in-depth expertise in the underlying methods.

To achieve this objective, we have developed femOHL (Finite Element Method for Overhead Lines), a numerical simulation tool built upon the FEniCSx framework. The code relies on advanced finite element formulations to capture the geometric and material nonlinearities arising from the truss-like configurations of pylons, conductors and foundations. It incorporates component-specific constitutive behaviours and accounts for environmental effects such as wind loading, temperature variations, corrosion-induced degradation and soil-structure interaction. Both static and dynamic solvers are implemented, enabling the simulation of progressive failure mechanisms that may ultimately lead to cascading collapse of pylons.

Developed within an industrial context, femOHL incorporates operational constraints and engineering rules to ensure consistency with real-world practices. By enabling the evaluation of structural behaviour under realistic environmental and operational conditions, femOHL supports the quantification of safety margins and the identification of critical configurations. Its modular architecture promotes extensibility and maintainability, allowing adaptation to a wide range of study scenarios. By providing a robust and scalable computational framework, femOHL supports TSOs in anticipating risks and strengthening infrastructure resilience.

Multiphysics Eddy-Current Simulation for Efficient Hybrid-Electric Aircraft Propulsion

Arshad Fasiludeen, Garth Wells

University of Cambridge

Submission type: Presentation

Presenter: Arshad Fasiludeen (University of Cambridge)

The aviation industry is under increasing pressure to enhance fuel efficiency and reduce carbon emissions, driven by the growing demand for global air travel. A promising approach to addressing these challenges is the development of hybrid-electric and electric flight technologies that utilise permanent-magnet synchronous motors to propel aircraft. However, optimising the electromagnetic performance of these systems requires efficient simulations of eddy-current phenomena. A challenging aspect of eddy current formulations is the large null space associated with the curl operator as well as discontinuous material parameters, which further complicate solver performance. We explore work incorporating coupling electromagnetics with thermal effects and fluid-based cooling methods. The results are compared across suitable geometries, emphasising the impact of these coupled effects on solver performance. This analysis again focuses on convergence rates and computational efficiency, aiming to optimise overall system performance.

L-PBF Processes via FEniCS: A Finite Element Framework for Heat Modelling and Accelerated Predictive Workflows

José Ángel Bejarano Vázquez

Isaac Toda Caraballo

Submission type: Presentation

Presenter: José Ángel Bejarano Vázquez (CENIM-CSIC, Universidad Complutense de Madrid (UCM))

In the scope of understanding the factors controlling the defects that commonly appear in the microstructures of materials produced by Additive Manufacturing (AM), a first step is the precise prediction of the thermal history. This work focuses on the development of a robust Finite Element Method (FEM) framework using FEniCS to simulate Laser Powder Bed Fusion (L-PBF) processes. The core of the numerical model consists of a moving heat source that traverses a discretized mesh. To capture the underlying physics as accurately as possible, the material properties across the mesh are dynamically modified throughout the simulation, accounting for phase changes and the evolution from powder to solid state. This FEM approach in FEniCS provides high-fidelity results and it has been experimentally validated. Nevertheless, its computational cost for large-scale optimization remains a challenge. Consequently, this study develops a methodology to accelerate these predictions. A key advantage of using FEniCS is its native Python integration, which makes the workflow for introducing Constrained Random Walks (CRW) extremely straightforward. These CRWs were used to efficiently generate a diverse dataset of scanning strategies and printing geometries within the FEM solver. Finally, this dataset was used to evaluate several Machine Learning (ML) models, such as Extremely Randomized Trees (ERT) and Feedforward Neural Networks (FNN), as well as Long Short-Term Memory (LSTM) networks for temporal profiles. The results demonstrate that while the physics are grounded in the FEniCS FEM simulations, the resulting surrogate models offer quasi-instantaneous predictions with minimal error, enabling the exploration of a wide variety of printing parameters and strategies.

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